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Sustainability in Focus: Exploring ESG performance and Financial Performance in SMEs vs large firms in the United Kingdom

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Acknowledgment

The thesis was written in the spring of 2024 at the Norwegian University of Life Sciences (NMBU) as a part of the Master of Science in Applied Economics and Sustainability.

I find this topic particularly interesting due to the global climate challenges. I hope this thesis will shed light on the dynamic relationship between sustainability investing and corporate financial performance.

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Abstract

This study analyzes the relationship between Environmental, Social, and Governance (ESG) performance and Corporate Financial Performance (CFP), focusing on Small-Medium Enterprises (SMEs) compared to large firms. The study contributes to existing knowledge on ESG and firm's profitability by delving into sustainability in SMEs. Using panel regression analysis on a sample of 355 listed United Kingdom (UK) firms from 2018 to 2022, the study explores how ESG practices influenced accounting-based profitability measurement - Return on Assets (ROA). The study examined ESG combined scores and their disaggregated components for a dynamic analysis. Moreover, corrected standard errors are used to address the issue of cross-correlations according to panel data.

The empirical findings revealed that the association between ESG performance and CFP varies depending on the analysis periods and firms' ownership structure. While no positive association was observed between ESG and CFP for all firms, a significant convex relationship was identified in the case of SMEs. The study attributes its findings to the volatility of the COVID-19 pandemic, emphasizing that firms prioritize cash flow towards maintaining business operations over investing in ESG activities. The nonlinear relationship suggests that SMEs face significant investment costs when implementing ESG practices. A straightforward implication for the study indicates that decision-making regarding ESG investment in the UK during uncertain times should be carefully planned to avoid potential profitability trade-offs. Furthermore, the fact that corporate governance is the critical driver in the ESG–CFP relationship suggests that investing in ESG activities should be directed to this component.

Keywords: ESG performance, financial performance, Return on Assets (ROA), the UK.

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List of Abbreviations

BEP	Basic Earning Power ratio
CBAM	Carbon Border Adjustment Mechanism
CEP	Corporate Environmental Performance
CFP	Corporate Financial Performance
CGP	Corporate Governance Performance
CSP	Corporate Social Performance
CSR	Corporate Social Responsibility
E	Environmental element
ESG	Environment, Social, and Governance
EU	European Union
FSB	Financial Stability Board
G	Governance element
ICB	Industry Classification Benchmark
LM	Lagrange Multiplier test
LSEG	London Stock Exchange Group
NFR	Non-financial reporting
OECD	Organization for Economic and Co-operation and Development
OLS	Ordinary Least Squares
RBV	Resourced-Based View
ROA	Return on Assets
SMEs	Small-Medium Enterprises
S	Social element

TCFD	Task Force on Climate-related Financial Disclosure
UK	United Kingdom
VIF	Variance Inflation Factor

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1. Introduction

Climate change is a challenging issue with far-reaching implications for the surrounding environment, society, and the global economy. Climate change's consequences, such as floods, droughts, and rising sea levels, are already being seen across the globe (Mimura, 2013). It is becoming increasingly urgent to mitigate and adapt to climate change through collaboration. As climate change necessitates cooperation efforts for mitigation, businesses play a critical role in reshaping the path towards sustainable development. When consumers increasingly become more aware of unethical business practices and shift their demand toward more sustainable products, acknowledging the impact of ESG initiatives on financial outcomes becomes imperative (Cini & Ricci, 2018).

Sustainable development and the term Corporate Social Responsibility (CSR) have been positioned since the 1950s. However, CSR became more codified into business practices as sustainability initiatives in the 1990s (Lee & Carroll, 2011). It emerged after the first United Nations conference in 1972 about environmental issues and grew even more after the Brundtland Commission defined sustainable development in 1987 (Latapí Agudelo et al., 2019). CSR term evolved to overcome the “Triple Bottom Line,” which embodies the idea that businesses have social and environmental responsibilities as important as their economic ones (Mendes et al., 2023). This idea contradicts “pure” capitalism from economist Friedman (1970), who argued that the business's only responsibility is to maximize profitability. While CSR initiated the firm's duties towards the environment and society, ESG was developed as a framework and guidelines for incorporating sustainable practices and accessing businesses' non-financial reporting (NFR) (Gillan et al., 2021; Hoang, 2018).

To delineate the research area, the UK is essential for investigating the relationship between ESG performance and CFP in SMEs. The case of the UK is interesting, considering its strategic importance in global business and commitment to sustainability (GOV.UK,

2021), which makes it an ideal context to explore the relationship. As we delve into this inquiry, the research aims to address specific research gaps. The favorable association between ESG – and CFP is supported by many studies (Margolis & Walsh, 2003; Whelan et al., 2021). However, most of these studies tested the relationship in large firms with broadly diverse ownership. SMEs tend to have a narrower and more concentrated ownership structure, resulting in a lower ESG disclosure level (Zadeh & Eskandari, 2012). The link between ESG-CFP in SMEs has not yet been vastly explored (Whelan et al., 2021), despite some of the studies that have tended to examine this relationship (Drempetic et al., 2020; Gholami et al., 2022).

Moreover, insufficient research has yet to analyze the ESG-CFP relationship separately for large firms and SMEs in the UK. This gap exists despite several studies highlighting the critical moderating role of firm size (Ahmad et al., 2021; Orlitzky et al., 2003). Therefore, linking ESG to SMEs' performance in the UK is crucially important, as we argue that the social contract between a firm and its stakeholders is equally vital for large vs small firms (Gholami et al., 2022).

From this perspective, this study investigates the impact of ESG performance on CFP in SMEs within the UK context. This study uses the panel data of 355 public UK-based firms in 8 industries (except financial, real estate, and energy) between 2018 and 2022. Data were collected from the ASSET4 database of Thompson Reuter DataStream. The study utilized the panel data technique, specifically the fixed-effect model, to study the effect of ESG performance on firms' profitability.

By delving into this intersection between ESG and CFP, this study aims to assess whether investing in ESG must come at a cost or if there is a potential for a win-win situation. This win-win scenario ensures that firms implementing sustainable solutions

contribute to societal well-being and enhance their financial performance. Taken together, this study explicitly explores and addresses these gaps by asking two following questions:

Research Question 1: *How does ESG performance impact firms' profitability?*

Research Question 2: *How does the correlation between ESG and financial performance differ between SMEs and large firms?*

1.1 Motivation and Objective

The overall prime motivation for this study is based on the work of Gholami et al. (2022), a unique study examining the relationship between ESG-CFP in SMEs in Australia. Their study improved prior studies by considering time and temporal lags and utilizing dynamic panel data analysis. By employing panel data, which shifts the focus from one individual at multiple times to multiple individuals at various time intervals, their study represented a notable advancement and enhanced the depth of the analysis.

Building upon the work of Gholami et al. (2022), this study expands our understanding by shifting the geographical focus from Australian to UK-based firms between 2018 and 2022. The UK and Australia are market-based economies, meaning larger listed firms have a more diverse ownership base than SMEs. This dynamic may influence the level of disclosure in sustainability practices, particularly among SMEs. By utilizing UK-based firms, the research aims to expand knowledge of how these ownership dynamics influence sustainability practices and a firm's profitability.

This study has two main objectives. Firstly, it aims to comprehensively examine how ESG practices enhance financial performance in UK firms. Moreover, the study also seeks to investigate the relationship between the disaggregated ESG components on firm profitability separately and determine which elements strongly impact CFP. Secondly, the study aims to explore the ESG-CFP relationship within SMEs compared to their larger counterparts, given

that larger firms have more capital resources and familiarity with sustainability reporting. These unique assets may incentivize larger firms to implement ESG practices.

From this point of view, the research highlights the crucial role of policymakers in recognizing obstacles and challenges SMEs face when integrating ESG criteria. Therefore, policymakers should create national incentives to support and facilitate ESG integration in SMEs. The study aims to make meaningful contributions to helping SMEs deal with challenges and guiding the UK towards sustainable development by providing policy implications.

1.2 Limitation

The limitation of this study is mainly associated with data availability and the research period. The first limitation lies in the restricted data availability. Initially, ten years were chosen when collecting firms' ESG data, particularly from 2013 to 2022. However, more than the first half of ESG data is incomplete or missing, and only data from 2018 to 2022 are mostly available for all firms. As the research aims to avoid missing observations as much as possible, the sample has been narrowed down to include only firms with available ESG data. On top of this, the data specifically include UK-based firms; thus, the results cannot be representative of other markets, such as the United States, due to different ESG regulations. Moreover, various metrics have been used to assess firms' financial performance in the literature, such as ROA, Tobin's Q, market value, and earnings per share (Ahmad et al., 2021), and these metrics might have influenced the models and the results differently. However, given time restriction and data unavailability, this study utilized a firm's ROA to assess its financial performance, which has also been used extensively by various researchers (Gholami et al., 2022; Mardini, 2022; Qureshi et al., 2021).

Secondly, the period chosen for the sample, specifically from 2018 to 2022, is highly volatile due to the COVID-19 pandemic. National restrictions and lockdowns may have led

corporations to behave differently, significantly influencing the relationship between sustainability performance and firms' profitability (Zhang, 2022). This volatility represents a potential limitation to the study, as it affects its uncertainty and generalization of the relationship between ESG and CFP. By contrast, a great strength of this study is that it assesses the disaggregated elements of ESG score on a firm's profitability and explores a potential non-linear relationship between ESG-CFP. These factors contribute to an understanding of their dynamic relationship.

1.3 Outline of the thesis

The thesis structure is described as follows. Section [2](#) presents background information regarding CSR, ESG performance, NFR, and the UK's ESG regulations. Section [3](#) provides theoretical background regarding the motivation of ESG initiatives, the crucial role of corporate governance through agency theory, the role of the institutional environment, and firms' competitive advantages to sustain financial performance. Section [4](#) presents the literature review and hypothesis development. Section [5](#) describes the sample, data collection, and variable selection. Section [6](#) explains the research's methodology and diagnostic tests for the models. Section [7](#) presents and evaluates the results, followed by the conclusion section in Section [8](#), which discusses and concludes the study's findings, its limitations, policy implications, and implications for future research.

2. Background

CSR traditionally emerged as a concept that focuses on corporate activities being more socially responsible. This concept involves the idea that organizations operating in this society should be accountable for their impact (Edmondson, 2022). For example, CSR focuses on corporate volunteering activities such as lowering their carbon footprint or engaging with charities. ESG, on the other hand, refers to how corporations integrate environmental, social, and government practices into their business models. ESG provides a more quantitative measurement of a business's sustainability, while CSR holds firms accountable for their social impacts in a qualitative manner. According to Gillan et al. (2021), the main difference between CSR and ESG concepts is that ESG includes government elements explicitly, whereas CSR includes governance issues implicitly. CSR governance issues are related to environmental and social problems; thus, CSR practices tend to be self-regulated and vary significantly. Therefore, ESG tends to be a more detailed term than CSR.

2.1 Environmental, Social, and Governance (ESG)

2.1.1 Definition

ESG stands for Environment, Social, and Governance, a concept first introduced and gained recognition among investors and businesses in the 2006 United Nations Principles for Responsible Investment (United Nations, 2006). ESG encompasses factors that reflect the company's commitment to the environment and society (Neri, 2021). Firstly, by considering the Environmental element (E), firms address their operational impact on the surrounding ecosystem and resource consumption. The element concerns the firm's pollution management, energy sources, CO2 emissions, and the strategy to combat climate change. Secondly, the Social element (S) covers the relationship between the companies and society. This element suggests that firms should pay more attention to labor practices, diverse cultures, community engagement, and human rights. Consumers nowadays are shifting their

focus towards cleaner products and responsible business for the sake of the environment (World Economic Forum, 2021). Understanding this trend and considering these social factors ensure that firms are on the right track to demonstrate their commitment to responsible business. Lastly, the Governance element (G) assesses the company's internal structure, including the firm's accountability, transparency, and management practices (Khan, 2011). These factors provide insights into a company's processes that help guide its decision-making, which is critically important in implementing ESG practices. Overall, these ESG dimensions provide a sustainability framework to evaluate a company's impact on the world, and they go beyond traditional financial metrics.

2.1.2 ESG practices drive financial performance

Companies integrate ESG factors into their business strategy through different channels. They come from setting long-term goals and objectives aligned with sustainable practices (Kalia, 2021). These goals and objectives include assessing ESG impacts on the company's risk profile, making operational changes to reduce carbon footprint and minimize cost, enhancing the engaged conversation with various stakeholders, and receiving feedback on the companies' ESG concerns. ESG criteria encourage businesses to consider their operational impact, not only due to ethical reasons. However, they can also lead to better financial performance through risk management, enhanced reputation, and ease of access to capital.

Firstly, ESG practices are considered a risk management tool for the firm because they help firms identify, assess, and mitigate various ESG-related risks. ESG issues are expressed through different channels, which can significantly impact the company's long-term overall performance. These risks include climate risks (physical risks), social risks (human rights and welfare), and governance risks (corruption). Climate risk represents catastrophic weather events that result in infrastructure damage or supply chain disruption.

Social and governance risks damage corporate reputations and increase legal and regulatory risks. Therefore, identifying these risks and responding proactively to ESG risks enables firms to protect their businesses during economic instability and enhance resilience.

In contrast, companies not addressing ESG risks are more likely to face reputational damage and higher financial liability. Proactive risk management helps businesses manage environmental and social-related risks, reducing potential financial liability (Whelan et al., 2021; Yankovskaya et al., 2022). Hence, the likelihood of costly incidents and lawsuits is decreased significantly.

Secondly, firms illustrating their commitment to the environment tend to yield advantages for reputation and higher customer demand (Porter & Kramer, 2006). Environmental responsibility serves as a unique proposition that differentiates firms from their competitors, which helps firms gain trust and loyalty from customers. The ESG criteria enhance the firm's reputation by showing its commitment to the environment and stakeholders. For example, ESG creates incentives to shift more focus to an employee's perspective than profitability, such as well-being, diversity, and fair labor practices. This shift creates a positive corporate culture and helps firms attract top talents in the long run. In other words, firm value can be enhanced when companies integrate ESG practices into their managerial strategies (Rezaee, 2016). Investors and lenders are increasingly aware of sustainability issues (Warren, 2020). Financial analysts also use the ESG framework to evaluate companies' non-financial performance and long-term resilience (Kalia, 2021). As a result, these stakeholders are more likely to support and invest in businesses that demonstrate an excellent commitment to their environment and society. Consequently, firms with higher ESG scores are more inclined to attract capital on favorable terms and gain a competitive advantage from a lower cost of borrowing. Overall, a firm positive reputation enhances its brand image towards stakeholders and quickly gains access to capital.

While the favorable relationship between ESG-CFP is well-documented, there is a notable critique about investing in ESG activities, potentially resulting in poor financial performance. Some argue that implementing ESG practices requires more upfront costs for companies, potentially impacting their short-term financial performance (Larcker et al., 2022). These costs include investing in green technology through Research & Development, training and educating employees, compliance, and reporting costs. The benefit of implementing ESG criteria takes time to materialize (*The Sustainability Edge: Business Benefits of Embracing ESG*, 2022); therefore, companies may face such an initial financial challenge. For these reasons, SMEs find it challenging to adopt ESG practices as their business case is entirely different from that of their larger counterparts (Drempetic et al., 2020). Managers in such firms may feel that they need more time, knowledge, and the capacity to tackle the climate crisis. Despite these challenges, recognizing the long-term benefits of implementing ESG criteria on firms' financial performance is crucial for enhancing sustainable business practices.

2.1.3 ESG performance & Non-financial reporting

Companies implement ESG practices through sustainability reporting alongside traditional financial reports to align their goals and vision and become socially responsible citizens. The ESG report, or NFR, contains qualitative and quantitative information about a company's ESG strategies and how they are implemented into a business's operation to affect outcomes (Daub, 2007). The NFR has become essential for many companies because the information conveyed from the report determines the accountability and transparency of firms' sustainability goals to investors and financial analysts. This report serves three main functions. Firstly, it becomes a primary channel for transparent communication with shareholders and stakeholders (Ernst&Young, 2014). It signals a firm's environmental and social performance and differentiates itself from less sustainable businesses to gain social

trust. Secondly, NFR reduces information asymmetry between managers and shareholders (Eccles & Krzus, 2010) because it addresses the concern of non-transparent corporate disclosure. By reducing information mismatch, NFR enhances investor's confidence and increases the firm's value. Finally, NFR is a handy strategic tool for sustainability plans (Gond & Herrbach, 2006). It controls and manages ESG-related risks for the firms and supports the opportunity and risk identification process. In summary, the NFR process requires internal strategic planning, such as identifying and managing risks and effective external communication. NFR has become a critical management tool for corporations interested in addressing climate change.

2.2 The UK institutional environment

Given the significantly important role ESG performance plays in a firm's financial performance, the UK is one of the players that has committed its journey to climate change mitigation. The case of the UK is interesting for several reasons: firstly, it left the European Union (EU) in 2020 and needs to establish its target to stay in line with global climate initiatives (Morgan Lewis, 2023); second, the introduction of the Carbon Border Adjustment Mechanism (CBAM) by the EU could pressure UK exporters in reporting embedded emission data in the upcoming years (Burke et al., 2021; Newham & Harrowell, 2023). This scheme has been in a transitional phase since October 1st, 2023, with mandatory verification of emission data required from January 1st, 2026, for the EU importers to meet the CBAM regulation. UK businesses maintaining their operation with the EU must provide detailed and robust data to meet the scheme's rules. False or misleading emission data could result in increased costs for the exporters, according to Newham & Harrowell from KPMG (2023), as calculating the EU's CBAM certification will have to adopt some default assumptions, which potentially lead to penalties. Hence, sustainability reporting is crucial for UK businesses to

comply with national regulations and maintain international competition. This discussion sets the stage for evaluating the evolution of the UK's ESG regulation post-Brexit.

Since departing from the EU, the development of ESG regulation in the UK has witnessed a dynamic change. Before Brexit, the UK played an important role in shaping the EU's ESG directives. After Brexit, it remained in a proactive position to further enhance its commitment to tackling climate change. Building upon its commitment to climate change, the UK introduced the Green Finance Strategy in 2019 to align its private sector financial movements with sustainable growth (GOV.UK, 2023b). This strategy underscores the importance of integrating ESG initiatives into investment decision-making and promoting transparency in sustainable finance. Simultaneously, the Streamlined Energy and Carbon Reporting policy was introduced and implemented, with efforts to promote transparency and encourage cost savings and emission reduction (GOV.UK, 2024). The policy required specific companies, such as large listed and unlisted companies and limited liability partnerships, to disclose energy use, carbon footprint, and greenhouse gas emissions alongside their annual financial report.

In 2020, the UK government incentivized to bring about the “green industrial revolution” to stimulate economic recovery from the Covid-19 pandemic. The UK government intentionally invested in making the UK a global leader in green technologies. The green revolution intends to: firstly, stay following with international initiatives, such as the Task Force on Climate-related Financial Disclosures (TCFD) established by the Financial Stability Board (FSB) (2015); and secondly, introduce a voluntary structure for businesses to report climate-related financial information nationwide by 2025 (GOV.UK, 2020). While the policy hopes to promote, standardize, and simplify sustainability reporting for companies, a limitation arises concerning its application. Most ESG regulations, including the TCFD, demand companies with over 500 employees to report (Jesus, 2023). As a result, small and

medium (SME) companies are not subject to the same climate-related disclosures and ESG regulations since they have less than 500 employees and 500£ million annual turnovers. Despite the commitment to prepare for the “green taxonomy” demonstrating the UK’s incentive to stay in line with international standards, these ESG regulations raise questions about the inclusivity of SMEs in addressing climate change.

According to the Organization for Economic and Co-operation and Development (OECD) (2019), SMEs represent roughly 99.7% and 60% of total employment among all OECD nations. They are the driving force of our global economy. Precisely, SMEs also form the backbone of the UK’s economy, which accounts for 99% of all businesses in the UK (GOV.UK, 2023a). SMEs contribute significantly to the country's economic growth and innovative approach by offering responsive customer service and fantastic employment opportunities. Given the large proportion of SMEs contributing to global economic growth, their impact has led to considerable loss of environmental quality (Moursellas et al., 2023). Studies indicate that SMEs account for approximately 64% of environmental impact in the EU, specifically industrial pollution (Constantinos et al., 2010). Given SMEs’ significant economic role and ecological impact on the UK economy, it becomes questionable whether the UK’s ESG regulation fosters or hinders the implementation of ESG practices in small and medium-sized businesses. According to Capterra (2023), 48% of British SMEs interested in pursuing ESG practices need more information on acting. Furthermore, 44% of managers and executives from these SMEs said that regulatory and insurance costs from implementing ESG threaten their operations. This information emphasizes potential barriers that SMEs face when integrating ESG and sheds light on improvement in the regulatory areas to ensure broader participation in sustainable initiatives.

3. Theoretical Background & Framework

This section presents comprehensive frameworks for understanding the motivation of ESG initiatives, how corporate governance is the primary driver of a firm's sustainability, the interlinked relationship between a firm's competitiveness and financial performance, and the fundamental role of the institutional environment in promoting and supporting ESG implementation.

3.1 Shareholder and Stakeholder Theory

Shareholders are people, individuals, companies, or even organizations that own shares in a private or public company, aiming to increase the company's share price (Ernst & Haar, 2021, p. 41). Shareholders are a subset of the larger stakeholder group of the company as they are also considered stakeholders. However, stakeholders are not necessarily the company's shareholders. They can buy and sell shares, receive dividends, have certain rights as company owners, and vote on strategic decisions. Because of this, their interest is for the company's management to develop strategies and implement activities that increase the profitability and financial success of the company. Most importantly, the shareholders' relationship with a company focuses on profit maximization (Ernst & Haar, 2021, p. 43). Therefore, in some cases, bad decision-making resulting in profit loss can cause shareholders to sell off their shares.

On the other hand, stakeholders are individuals or organizations impacted by a company's operating activities or outcomes. These individuals and organizations can be internal (shareholders, managers, and employees) or external (vendors, suppliers, customers, and the community). In contrast to shareholders, stakeholders tend to be more interested in the company's longevity and success (Ernst & Haar, 2021, p. 45). For example, for customers to buy high-quality products and receive good customer service, the company must invest in R&D and train its customer service staff. These investments might not necessarily increase

profitability in the short term but can offer a positive outlook in the long run. Long-term performance ensures decent wages for employees and timely payments for goods and services received from suppliers.

The term “stakeholders” relating to organizations can be traced back to its origin at the Stanford Institute in 1963 when R. Edward Freeman published his foundational work in “Strategic Management: A Stakeholder Approach.” Freeman (1984) stated that companies should pay attention to the interests of all stakeholders, which is aligned with Ernst & Haar (2021, p. 44), instead of focusing only on shareholders. This approach creates value for everyone involved with the company’s operational activities and helps it achieve its long-term success. As interest in CSR increases, stakeholders’ theory has become more known as one of the foundation elements of the CSR movement. García-Sánchez and Noguera-Gámez (2017) suggested that the relationship with stakeholders determined the potential for firms to be profitable. Thus, companies might consider how their decision-making processes impact various stakeholders rather than making choices based only on the interest of shareowners. Overall, stakeholders’ theory was built on the premise that a company’s responsibility goes beyond maximizing its shareholders’ values. A business operating ecosystem contains various stakeholders, each contributing to the business's sustainability and ability to create value for the stakeholder group (Gutterman, 2023).

3.2 Agency Theory

In the realm of ESG performance, agency theory is concerned with aligning the interests between the principles (shareholders) and agents (company managers) to ensure effective decision-making and maximize shareholder value (Elmanizar et al., 2019; Singh et al., 2023). The decision-making process is distinct for large and SMEs due to their ownership structure. Large corporations often have dispersed ownership, where companies separate ownership and management. Therefore, decision-making is spread among many shareholders

or institutional investors through voting, and no individual has strong dominance. In contrast, SMEs have more concentrated ownership or duality leadership (Hakimah et al., 2019). Thus, a particular group of individuals has a uniquely strong influence on decision-making dynamics compared to larger firms. When there is disagreement between different parties, for instance, managers prioritize their interests over those of shareholders, and this conflict creates an agency problem. The theory highlights its relevance in understanding corporate decision-making dynamics between large and SMEs and its implication for ESG practices and CFP. It also emphasizes how effective corporate governance is crucial in mitigating the agency problem.

Corporate governance influences the decision-making process in large and SMEs, from designing the organization's structure to integrating ESG initiatives. It refers to board size, independent board members, executive compensation, shareholder rights, and financial reporting transparency. As Tang (2022) indicated, companies with weaker governance mechanisms have a higher probability of agency problems; therefore, they tend to perform worse. Thus, understanding the role of corporate governance through the lens of agency theory becomes essential, especially when examining whether the difference in a firm's structure fosters or hinders the alignment of sustainability initiatives with shareholders' interests. Large corporations have dispersed ownership; hence, information asymmetry - situations where the agents have more information than the principles, can quickly arise and lead to a conflict of interest. ESG disclosure may mitigate this conflict of interest by reducing information asymmetry (He et al., 2013; Kazemi & Rahmani, 2013). ESG initiatives require companies to disclose relevant information to enhance transparency with shareholders about how the companies are doing to manage their risks. Therefore, investing in ESG practices reduces the information gap between the agents and the principles, fosters a concrete engagement, and encourages trust with various stakeholders.

In owner-operated firms, the close relationship between the owners and the management represents a different set of agency problems. Ownership structure in SMEs is more concentrated and involves direct owners in daily operating activities. While the small company's structure facilitates a more direct alignment between the agents and the principles, it is questionable whether ESG considerations are integrated into the decision-making process. For example, SMEs operate with fewer employees and are closely tied to the owners; hence, financing for SMEs mostly depends on private equity (Hakimah et al., 2019) and does not require public accountability. Moreover, SMEs usually face resource constraints; therefore, investing in ESG initiatives can be challenging and less compelling (Dzomonda, 2022). Thus, the owner may find it easier to prioritize personal goals for short-term gains over the firm's long-term sustainability.

Nevertheless, the proximity of the owners and the management in SMEs also presents a potential for effective and straightforward communication through enhancing decision-making and reducing interest conflict (Fidanoski et al., 2014). For example, fewer layers of hierarchy and bureaucracy in a family-owned business allow for a more transparent discussion on sustainability goals and practices. A manager's acknowledgment of the importance of ESG initiatives may foster a culture of innovation and continuous improvement for the firm. Overall, SMEs can institute effective corporate governance to ensure a smooth process of managing the enterprises in the best interests of all stakeholders (Hakimah et al., 2019).

3.3 Institutional Theory

Institutional theory posits that a firm's behavior is influenced by institutional pressures such as regulations and societal norms. This institutional environment, defined as regulatory surroundings and social expectations (Scott, 2008), includes specific organizations such as the government, public opinion, and social media (Bansal, 2004). The theory

emphasizes the importance of the institutional environment in supporting and integrating ESG practices into a corporation's operational activities. Moreover, it highlights how institutional weaknesses affect the relationship between sustainable and financial performance. Low ESG practices occur because firms do not recognize the potential benefit of being socially responsible, hence prioritizing capital accumulation (Garcia, 2020).

In the context of institutional influence on corporate behavior, companies can adopt various strategies to comply with social norms and gain social trust from the institutional environment. This institutional climate, which includes regulatory bodies, industry peers, and socially aware customers, can significantly influence a firm's operations. One strategy is adopting ESG practices in the company's operational processes (Meyer & Rowan, 1977). By addressing ESG-related issues and implementing sustainable practices, companies can enhance their reputation and align themselves with social values. This helps them avoid reputational damage from undesirable business practices and improves their financial performance. According to Liu et al. (2022), the impact of ESG activities on a firm's financial performance is influenced by the mediating role of non-financial performance, which is determined by the level of institutional pressures from the government, consumers, and competitors. This perspective aligns with Pinheiro et al.'s (2023) research, which concluded that "companies engage more in ESG practices when they operate in countries with greater economic and international trade freedom." In other words, the country's legal system and national culture play a crucial role in mediating the impact of ESG performance on firm financial performance (DasGupta & Roy, 2023).

3.4 Resource-based view Theory

Resource-based View theory (RBV) provides a valuable aspect for assessing the impact of ESG performance on CFP in both large and SMEs. The theory emphasizes that each firm has unique internal resources that can be used for its competitive advantage. These

resources must be valuable, rare, inimitable, and non-substitutable and can be leveraged to exploit more opportunities for companies' business outcomes (Barney, 1991). Such resources, from tangible (factory, machinery) to intangible assets (human capital, capital structures, leadership), allow firms to conduct ESG practices to gain a public reputation. By integrating these unique resources into their core operational process, firms make it difficult for other competitors to replicate, hence consequently improving the firm's financial outcomes by meeting stakeholders' demands. While institutional pressure motivates firms to conduct CSR to gain social legitimacy, resource-based view theory distinguishes firms from others based on their sustained competitive advantages.

RBV underscores the importance of recognizing the nature of competitive advantages as they constantly change following the surrounding environment. The constantly changing environment requires firms to adjust their operational activities to adapt to new opportunities and challenges. This theory provides insights into how firms can align their competitive advantages with market opportunities to build a resilient foundation for success. Russo and Fouts (1997) described an example where a company faces two options to meet stakeholder's demand to reduce pollution. The first option is to install off-the-shelf filtering equipment only for compliance. The second option is to alter the design process to engage in pollution reduction actively. While both options address the stakeholder's demand, the RBV suggests that actively supporting pollution reduction in option two may provide firms with a more sustained competitive advantage as firms do not simply comply. Overall, the RBV theory helps explain why some firms outperform their peers and constantly acquire abundant resources as they continue to achieve financial outcomes (Huang & Yu, 2024).

4. Literature Review

4.1 ESG-CFP relationship

4.1.1 Positive findings

Meta-analysis studies have long examined the association between ESG and a firm's profitability. Friede et al. (2015) reviewed more than 2200 studies before 2015. They found a consistently positive relationship between ESG performance and operational efficiency, stock performance, and lower cost of capital across developed markets. They found that roughly 90% of all the studies, including corporate focus, found a non-negative relationship between ESG and CFP. This non-negative relationship suggests strong evidence for the positive impact of ESG investing in the business case, and the relationship between ESG - CFP remains stable over time. Another meta-analysis, which utilized 1000 articles published between 2015-2020, also found this favorable relationship (Whelan et al., 2021). They took a different approach by dividing these 1000 articles published in this period into two main categories, compared to the first meta-analysis, which first employed results from "vote-count studies." Then, they aggregated for a second-order "meta-analysis." The first category focused on Corporate Financial Performance, using operating metrics such as ROA, Return on Equity, and stock performance. The second category focused on Investment Performance from the investor's perspective, using investment metrics such as Alpha, Beta, and Sharpe ratio on a stock portfolio. The result continuously suggested a strong correlation between ESG-CFP, with 58% positive for "corporate" studies. Positive correlation explains that firms are more motivated to disclose ESG information to improve their reputation and accountability, consequently leading to higher financial performance (Dellaportas et al., 2012; Forcadell & Aracil, 2017). Investing in ESG initiatives, therefore, is considered a downside protection for corporates, especially during an economic crisis, due to the company's resilience to market volatility (Whelan et al., 2021).

4.1.2 Negative findings

Although most views of the ESG-CFP relationship are positive, there are neutral, mixed results in the literature. For investment studies that focused on risk-adjusted returns such as the Alpha or Sharpe ratio, the returns from investing in ESG initiatives were generally indistinct from traditional investing returns (Friede et al., 2015) or remained neutral (Whelan et al., 2021). This result is potentially the source of ESG-CFP misperception, where neoclassical economists and institutional investors view ESG-CFP relations as neutral (Fama, 1991; Markowitz, 1952). The neutral relation is consistent with Milton Friedman's (1970) argument that “the primary social responsibility of business is to maximize profits.”

Nirino et al. (2021) proved a contradicting view between corporate controversies and a company's financial performance using ESG practices as the moderating role. The study utilized a linear regression model to analyze 356 listed European companies and confirmed a significantly negative relationship between corporate controversies and financial performance. They found that ESG principles should not be used to mitigate the negative impacts of controversies but rather to avoid them, despite the increasingly important role of ESG practices in business strategies. This view contradicts Whelan et al. (2021), who claimed that ESG investing is a downside protection for firms during bad times. The idea behind Nirino et al.'s (2021) research was that little is known about how corporate controversies related to the impact of CSR on firm financial performance. More importantly, no compelling evidence supports the positive effect of ESG practices on these controversies – profitability relationship. In other words, even when companies invest in ESG activities, there is no guarantee that financial performance will see improvement in the presence of controversies.

Given insights from prior literature, this study intends to explore the association between firms' overall ESG performance and profitability. The first hypothesis is proposed below:

Hypothesis 1 (H1). *A firm's ESG performance is positively correlated with its financial performance.*

4.2 Firm Size and Financial Performance

Another body of literature looked at a diverse correlation of the impact of ESG performance on financial performance through the lens of firm size moderating role. Firm size is related to CFP because the trade-off between corporate ESG performance disclosure and profitability may vary depending on the firm's characteristics (Cormier & Magnan, 1999). A firm's characteristics, such as its size, board composition, reputation, and market exposure, are distinguishable between large and SMEs; therefore, they should be considered when analyzing ESG-CFP. This view also aligns with Ahmad et al. (2021), who concluded that firm size determined the relationship between ESG and CFP of the FTSE350 UK firms.

Some studies suggested that firm size positively moderates the ESG-CFP based on two main arguments (Sánchez-Infante Hernández et al., 2020; Wang et al., 2018). Firstly, according to the resource-based view theory, large firms are more willing to invest in ESG activities than SMEs because they can access more available resources and capabilities (Johnson & Greening, 2017). SMEs cannot engage intensively in ESG practices as they use available funds to support their core operations and enhance their competitive edge. Larger firms tend to integrate sustainability initiatives at a much larger scale than smaller firms (Cornett et al., 2016), as higher costs through ESG implementation are much more achievable by larger firms. Larger firms also have more access to data availability, as indicated by Cornett et al. (2016); therefore, disclosing ESG performance may result in more significant competitive advantages and consequently lead to higher profitability.

Secondly, institutional theory suggests that corporate institutional environments are influenced by other stakeholders, such as the government, society organizations, non-governmental organizations, the media, investors, consumers, and employees. These

stakeholders pressure corporations, leading them to adopt CSR/ESG activities to respect institutional regulation. However, judgment from not participating in ESG practices is more likely to affect visible firms due to reputation effects (Udayasankar, 2008). Therefore, less visible firms like SMEs tend to engage less in CSR/ESG activities than visible and large firms.

In contrast, prior studies suggest that firm size negatively moderates the ESG-CFP relationship (Ting, 2021). Some claimed that SMEs have an advantage in investing in ESG practices because the marginal utility of enhanced legitimacy could be higher for these less visible firms (Udayasankar, 2008). Therefore, ESG initiatives motivate these SMEs to gain legitimacy and reputation. Moreover, Baumann-Pauly et al. (2013) and Wickert et al. (2016) argued that SMEs cannot report and communicate these advances externally despite having several characteristics that can positively influence their ESG investing and profitability. The discussion about SMEs' communication gap is consistent with Dremptec et al. (2020). Their study criticized the computation of ESG scores. They argued that larger firms have more resources, thus producing higher-quality ESG data for rating agencies. Do larger firms exhibit greater sustainability than smaller firms, or do they have better sustainability reporting?

Given the state of existing literature, there are mixed results in determining the ESG-CFP relationship of SME firms compared to large firms, and limited studies have been reported within the context of SMEs (Dremptec et al., 2020; Gholami et al., 2022). Therefore, more research needs to be conducted on the influence of ESG performance on the financial performance of SMEs. Thus, the second hypothesis is:

Hypothesis 2 (H2). *The relationship between ESG performance and CFP differs for large firms compared to SMEs.*

4.3 ESG components

For the individual effect of E, S, and G on CFP, Friede et al. (2015) found that E and G demonstrate a moderately more positive correlation than S-studies, consistent with Dalton et al.'s (1999) findings. The positive correlation between G and CFP exhibits more than 62.3% of all cases in the sample of vote-count studies. Friede et al. (2015) argued that corporate governance is the main driver for a firm's sustainability, where a higher level of ESG disclosure creates a competitive advantage for the firm and consequently increases its financial performance over time. However, they also demonstrated that G aspects have the highest negative correlation (9.2%) compared to E (4.3%) and S (5.1%), again indicating the strong impact of G aspects on CFP. This meta-analysis shed light on further assessing the specific sub-criteria of ESG on CFP.

Regarding the relationship between corporate environmental performance (CEP) and CFP, Moneva and Ortas' s (2010) findings supported the positive linkage. The paper utilized a partial least squares model, which was used for the first time in the field, to analyze 230 European firms. They claimed that corporations with higher levels of environmental performance will demonstrate better financial performance levels in the future. However, according to Lankoski (2000), the debate between a firm's environmental and economic performance should not be framed as whether or not but when the improvement in CEP results in improved CFP or a so-called win-win situation. The result showed that this relationship is not constant but an inverted U-shaped function. The concave function means a win-win situation exists if firms are in the rising part of the environmental profit curve. After the turning point, firms no longer benefit from environmental performance improvements.

Studies show conflicting results regarding the association between corporate social performance (CSP) and CFP. According to Lu et al. (2014), the positive relation between the specific CSR and CFP is confirmed in some studies (Margolis & Walsh, 2003; Orlitzky et al.,

2003). However, this relationship is not static; it constantly changes over time. The study reviewed 84 empirical studies using mixed methods of content analyses and statistical analyses from 2002 to 2011. Despite the positive link, they concluded that this relationship varies depending on a specific background, such as a particular industry or country. Therefore, the CSP-CFP relationship is complex and remains inconclusive.

Many studies in literature supported the intuition that good governance leads to good profitability (Affes & Jarboui, 2023; Kyere & Ausloos, 2020); however, there has been insufficient conclusive evidence on this linkage over the years (Khan et al., 2018; Pande, 2011). A positive but insignificant correlation between corporate governance performance (CGP) and CFP was found in Van de Velde et al.'s (2005) findings, which was in line with Gompers et al.'s (2003) findings, where they confirmed that more robust governance increases firm's value and consequently firm's profit. However, Selvaggi and Upton (2008) strongly emphasized that improved corporate performance causes enhanced corporate governance, not vice versa. Ineffective corporate governance may harm firms' performance, possibly due to excessive leverage (Kijkasiwat et al., 2022) or short-term board diversity, as it disrupts group cohesiveness and communication (Khan et al., 2018). In a financial crisis, higher leverage levels challenge a firm's ability to meet debt obligations, maintain liquidity, or access additional finance. Therefore, weak corporate governance exacerbates the negative impact on a firm's profitability.

Insights of the individual elements of ESG performance from the literature, together with the volatile period utilized in this research, provide ground for the second hypothesis to be developed as below:

Hypothesis 3 (H3). *Among the ESG performance individual elements, governance performance exhibits a stronger negative correlation with CFP than environmental and social performance.*

5. Data

This section presents ESG data and the firm's financial performance. It discusses sample selection, including the requirement for the dataset and variable selection of ESG and financial data. The section concludes by summarizing all variable notations used for the analysis.

5.1 Sample selection

The study sample comprises 1408 firm-year observations, including 355 active UK firms listed on the London Stock Exchange Group (LSEG) between 2018 and 2022. Initially, the period of 2013-2022 was collected for the analysis. However, given the significant lack of ESG data in the first five years, the final sample includes only the last five years. The list of these UK companies is obtained based on ASSET4 ESG data provided by Thomson Reuters DataStream. ESG data and financial information are also collected accordingly based on the list of companies with ASSET4 data from Thomson Reuters DataStream. This terminal provides comparable and in-depth coverage of over 7000 global companies (Thomson Reuters, 2019). The final number of firms is selected based on the following criteria:

- I. Constituents of companies from the UK with ASSET4 ESG data: LA4CTYGB (669)
- II. Equity, excluding Investment Trust (573)
- III. UK market + public firms (550)
- IV. Active firms (539) + GBP currency (525)
- V. All sectors, except financial, real estate, and energy (406)
- VI. Exclude firms with missing ESG data (355)

Firstly, a list of UK companies is identified in the ASSET4 ESG data. The UK was chosen because, as indicated in the literature review section, there is currently a lack of

research on ESG-CFP performance in SMEs in the country. The ASSET4 database provides various ways to collect sustainability information. One is to localize the companies with ESG data through constituent lists. LA4CTYGB gives a constituent list of 669 UK firms from various sectors, including investment trusts, which disclose ESG data (Copenhagen Business School, 2023).

Secondly, the number of firms on this list includes equity and investment trust. An investment trust is a public limited company that pools money from investors and aims to make money by investing in other firms (Smith, 2024). Due to this business nature, investment trusts were not included as the study focuses only on equity firms. By concentrating solely on the equity market, the research aims to delve deeper into the dynamic ESG-CFP performance and sustainability business strategies specific to companies—*screening result: 573.*

Thirdly, this research assesses only publicly traded companies from the LSEG, as public firms naturally have more data available than private companies. Moreover, these public firms have headquarters across different countries, so they must comply with ESG regulations depending on the firm's location. For the scope of the analysis, the sample includes only public companies in the UK market, given the significant institutional context, considering that these public firms form the same ESG regulations basis for comparison—*screening result: 550.*

Fourthly, around 2% of these firms were delisted or suspended from the LSEG in the past years, and 2% were firms that disclosed financial information in different currencies. Hence, it is necessary only to choose actively traded firms and firms with GBP currency. Similar currency makes the analysis more reliable and comparable between various firms—*screening result: 525.*

Fifthly, firms from various industries are included EXCEPT financial, real estate, and energy. The financial sector has specific characteristics, such as different capital structures and higher leverage, making it difficult to treat it as comparable. Therefore, unlike other industries, they must comply with different ESG regulations. The real estate industry, including investment trusts, often has unique business models and higher risk profiles. While this industry may have ESG impacts, such as energy consumption, community collaboration, and disclosure transparency, these factors tend to be less material and do not primarily drive the financial performance of the real estate industry. The energy sector was also excluded due to its more extensive accumulation of property, plants, and equipment relative to total sales and higher capital expenditures relative to total sales than other industries. This business nature makes it a potential outlier in the sample; therefore, the energy sector was not chosen. The financial, real estate and energy industries were excluded from the sample—*screening result: 406*.

Finally, firms with limited ESG data across five years were also excluded as they might not provide enough variations for the analysis. These firms have only disclosed one or two years of ESG data, possibly due to incomplete data reporting or being in the initial phase of ESG disclosure. The remaining firms have reported at least three years of ESG disclosure and have been chosen for the final sample, which results in 355 firms.

5.2 Variables selection

5.2.1 Independent variables

This study uses the firm's voluntary ESG performance disclosure as predictor variables. They are expressed as a score to measure a firm's ESG performance, commitment, and effectiveness towards sustainability (LSEG Data & Analytics, 2023). ESG information can be obtained from several databases, including ASSET4 and Bloomberg. Nevertheless, this study utilizes the ASSET4 from Thompson Reuters DataStream, which the Norwegian

University of Life Sciences provided. The ASSET4 measuring scale is based on 280 key performance indicators, including diverse E, S, and G elements. The database is used extensively by researchers (Ahmad et al., 2021; Gholami et al., 2022), and its ESG scores reflect a robust indicator of companies' ESG performance where biases towards company size and transparency are minimalized (Thomson Reuters, 2017).

ESG performance disclosure is a comprehensive framework encompassing three main pillars: E, S, and G. Each pillar is calculated by multiplying each category using companies' publicly reported data with weights varying across industries (Thomson Reuters, 2017). These categories, grouped into ten different ones, are displayed in Table 1 below. The regression model with each element is used distinctly to test the individual effect of the three components on CFP, providing a thorough analysis of the impact of ESG performance disclosure.

Table 1. 10 main categories to calculate ESG score

Environment	Social	Governance
Resource Use	Workforce	Management
Emission	Human right	Shareholders
Innovation	Community	CSR Strategy
	Product Responsibility	

Note: Sourced from LSEG Data & Analytics (2023).

While ESG individual pillar scores provide the ESG performance of the firm in each E, S, and G pillar, the ESG overall score is calculated by aggregating these three score pillars. The ESG overall score includes three leading indicators: ESG score, ESG combined score, and ESG controversies score. The main difference between the three indicators is that the ESG controversies score incorporates the impact of adverse media events in which companies were involved, penalizing firms by discounting their ESG combined score from the ESG score (LSEG Data & Analytics, 2023). When firms are not engaged in ESG controversies, their ESG combined score often equals the ESG score. For this reason, the ESG combined

score is preferred in this analysis as it reflects market bias, which large firms suffer from as they typically attract more public attention than SME firms. This score ranges from 0-100, with a higher score indicating a more substantial ESG rating (LSEG Data & Analytics, 2023).

Overall, to evaluate the impact of ESG performance on CFP, the one-year lag variables of ESG combined score (ESG_{t-1}), Environmental score (ENV_{t-1}), Social score (SOC_{t-1}), and Government score (GOV_{t-1}) are to be used as independent variables in the analysis. Lys et al. (2015) argued that the financial benefits of incorporating ESG activities represent a lag as they emerge in subsequent periods. Using lag variables allows time for the impact of ESG initiatives and practices to manifest in the data. It helps reduce noise or short-term volatility due to changes in reporting methodologies. For example, ESG scores from the previous year may potentially impact the firm's financial performance in the current year. Hence, including one-year lag variables smooths out the disturbance and provides a clearer picture of the effectiveness of companies' ESG efforts.

5.2.2 Dependent variable

Studies have been subjective regarding which measurement to assess a firm's financial performance. Some have employed the accounting-based measurement, ROA, as the main proxy for assessing firms' financial performance (Gholami et al., 2022; Giannopoulos et al., 2022) because it reflects the most accurate measure of accounting performance (Brooks & Oikonomou, 2018). In contrast, Ullmann (1985) argued that market-based measurement, such as Tobin's Q, is better for looking at a firm's ability to generate future profit rather than at past performance provided by ROA. Nevertheless, an accounting-based measurement is better for assessing organizational behaviors and capabilities (Orlitzky et al., 2003). ROA provides a straightforward insight into the company's operational effectiveness and can be significantly relevant to analyzing sustainability management

strategies. Hence, this paper will utilize accounting-based measurement as a proxy for evaluating large firms and SMEs' financial performance.

In financial terms, earning power considers the firm's ability to generate profits. Different earning power metrics can all be used to measure economic performance, such as ROA and Basic Earning Power (BEP). ROA indicates how well the company generates net income based on its invested assets. While the BEP ratio is similar to ROA, it considers the Earnings Before Interest and Tax (EBIT) and includes non-operating income such as dividends. For this reason, BEP is more efficient when comparing how well different companies generate revenue from their assets, as it disregards the tax and borrowing levels varying from one firm to another (Bloomenthal, 2023). Hence, this research will employ the BEP ratio as the main proxy for measuring a firm's financial performance, which is also used by Gholami et al. (2022) in their analysis. It is also noteworthy that the BEP ratio is often used as it provides a more accurate overview in company-to-company comparison, according to Bloomenthal (2023). For the convenience of the analysis, BEP, expressed in %, will be coded as ROA in the study and is defined as below:

$$BEP = \frac{EBIT}{TOTAL\ ASSET}$$

5.2.3 Control variables

Other firms' specific characteristics used by Gholami et al. (2022) are also included in the analysis as control variables. These variables are the log of total assets (LNTA), property, plant, and equipment ratio (PPE), annual revenue growth (GROWTH), cash ratio (CASH), and debt ratio (LEVERAGE). Including these variables helps control for different firms' specific effects, which could potentially impact their financial performance and further the ESG-CFP relationship.

Log of total assets (LNTA): is the main proxy for capturing firm size (Ahmad et al., 2021). Large firms typically have higher total assets than SMEs; therefore, they have a grander scale of operations and investment capacity. For this reason, large firms may be more closely related to stakeholders' expectations and concerns regarding the business's social responsibilities. Including total assets ensures a fair comparison between large firms and SMEs.

Property, plant, and equipment (PPE): represents the company's investment in tangible assets. PPE is expressed as the PPE/total sales ratio and indicates how efficiently the company generates sales based on the amount of PPE investment. Controlling PPE provides insights into the company's capital allocation and how it manages its physical assets in alignment with ESG considerations (Kao, 2023), such as energy consumption, workplace conditions, and governance practice. Large amounts of PPE during a volatile period signal inefficient capital allocation.

Debt ratio (LEVERAGE): represents a firm's unsystematic risk and is measured by total debt to total assets. Companies typically borrow capital to finance their investment and operational activities. In some cases, firms with a high level of sustainability acquire a lower level of debt cost, thus increasing their CFP (Orlitzky & Benjamin, 2001). However, companies with higher leverage levels also have higher financial risks in an economic downturn, which limits them from accessing additional funds for sustainability projects. In such a case, these firms may prioritize short-term financial gain over long-term ESG considerations.

Revenue growth (GROWTH): is an indicator of business performance. It is defined as an annual percentage change in sales over the year. Revenue growth reflects the company's capacity to invest in ESG initiatives. Higher revenue growth allows companies to allocate their financial resources towards ESG objectives.

Cash ratio (CASH): is expressed as cash/total assets and represents a company's liquidity. This ratio indicates a firm's flexibility in allocating resources for an investment or managing financial risk. Higher cash reserves enhance the capacity for capital expenditure; hence, firms are more likely to increase the incentive for ESG investments (King & Santor, 2008).

Industry (IND) and Year (YEAR): Dummy variables control the industry-specific and time-specific effects.

Table 2 below summarizes all the dependent, independent, and control variables and their notation definition.

Table 2. Summary of variables used in the analysis

Category	Notation	Definition	Expected sign
Dependent variable	$ROA_{i,t}$	Firm profitability (%) for company i at time t . Expressed as $\frac{EBIT}{TOTAL\ ASSET}$.	
Independent variables	$ESGC_{i,t-1}$	ESG combined score for company i at time $t-1$.	Positive (+)
	$ENV_{i,t-1}$	The environmental score for company i at time $t-1$. The score is derived from weighted scores on indicators including resource use, emission, and innovation.	Negative (-)
	$SOC_{i,t-1}$	The social score for company i at time $t-1$. This score is derived from a weighted score on workforce, human rights, community, and product responsibility indicators.	Negative (-)
	$GOV_{i,t-1}$	The governance score for company i at time $t-1$. This score is derived from weighted scores on indicators including management, shareholder, and CSR strategy.	Negative (-)
Control variables	$LNTA_{i,t}$	Natural logarithm of total assets for company i at time $t-1$.	Negative (-)
	$LEVERAGE_{i,t}$	$\frac{TOTAL\ DEBT}{TOTAL\ ASSET}$ ratio for company i at time $t-1$.	Negative (-)
	$PPE_{i,t}$	$\frac{PPE}{TOTAL\ SALES}$ ratio for company i at time $t-1$.	Negative (-)
	$GROWTH_{i,t}$	Annual revenue growth rate (%) for company i at time $t-1$. Expressed as: $\left(\frac{Net\ sales_t - Net\ sales_{t-1}}{Net\ sales_{t-1}} \right) \times 100$	Positive (+)
	$CASH_{i,t}$	$\frac{CASH}{TOTAL\ ASSET}$ ratio for company i at time $t-1$.	Positive (+)
	IND	Dummies for industry group	
	YEAR	Dummies for year	

Note: Notation of variables is used to perform analysis in Stata. ESG and financial information are extracted from the Thompson Reuter DataStream & ASSET4 Database.

6. Methodology

Generally, studies analyzing how ESG initiatives affect corporate financial performance typically use a panel of companies, as it considers variation in both cross-sectional and time-series dimensions (Whelan et al., 2021). The meta-analysis also indicates that investment studies, such as ESG funds and portfolios, usually use time-series data in their analysis. Panel data allows researchers to consider time and temporal lags in the analysis, minimizing estimation bias. Therefore, panel data analysis is employed in the research to study the relationship between ESG and CFP using the fixed-effect estimation model, which was confirmed by the Hausman test.

6.1 Methodology for classifying firm size

The study's methodology includes first segmenting the sample into two subsets of large and SMEs to test for the difference in the ESG-CFP relationship as hypothesized in H3. One could consider using a threshold for total assets or employee numbers to classify firm size provided by GOV.UK (2023a); however, this approach raises some challenges as it allows for an inadequate gap between the number of large and SMEs in the sample. For this reason, this study utilizes the approach Gholami et al. (2022) used, where they suggested using the dichotomy of LNTA based on the LNTA's mean. Using the mean of LNTA to dissect firms into large (large firms = 1) and SME-sized firms (SMEs = 0) provides a relative measure of firm size. This approach considers the distribution of firm size in the sample and ensures consistency in defining firm size. Hence, it adapts to the sample's variation characteristics and makes more meaningful comparisons across time.

Additionally, due to the significant fluctuation of firm total assets and the highly volatile period used in the analysis, some firms are classified as large and SMEs at different points in time. Therefore, the majority rule-of-thumb is applied to provide consistency in firm size classification. This rule indicates that the size classification depends on most periods in

which firms are large or SMEs. For example, if a firm is consistently classified as a large firm or SME for at least three years, it will continue to be classified as a large firm or SME respectively for five years.

6.2 Choosing Model Specifications

Before choosing the fixed-effect estimator approach for the analysis, several tests were performed to determine which model approach should be used to analyze the ESG-CFP relationship. The F-test for fixed effects compares a robust model with the pooled Ordinary Least Squares (OLS) to examine whether the goodness-of-fit has changed between the two models (Park, 2011). The null hypothesis indicates that the fixed effect is not significant. Therefore, rejecting the null hypothesis means the goodness-of-fit has significantly increased, and the fixed-effects model is better than OLS ([Appendix A](#)).

The Breusch-Pagan Lagrange Multiplier test (LM) examines the random effects in each regression model. The null hypothesis of the LM test implies that the variance of the random effect is equal to zero. Therefore, rejecting the null hypothesis means that random effects are present in the panel. Random effects suggest there may be unobserved individual-specific effects correlating with the dependent variables, which the model OLS does not account for; hence, OLS should not be used for the analysis (Park, 2011). The result from the LM test for random effect is presented in [Appendix B](#), where the random effect is significant in all models.

After checking for random effects, the Hausman test determines whether the fixed or random-effect models are preferable (Brooks, 2019, p. 537). The null hypothesis of the Hausman test indicates that the coefficient difference between random effects and fixed-effects models is zero. Hence, rejecting this null hypothesis means that the individual effects are significantly correlated with some of the regressors; thus, using the random effects model

is problematic (Park, 2011). All the regression results in [Appendix C](#) confirmed using the fixed-effects model as small p-values reject the null hypothesis.

6.3 Estimation models

6.3.1 Hypothesis 1

As indicated in Table 2, ROA (BEP) represents a proxy for a firm's profitability and provides a fair comparison between companies as it is not a market-sensitive indicator. The first equation regresses ROA against the firm's one-year lagged ESG combined score and other control variables for all firms regardless of size. This model provides an overview of the economic impact of investing in ESG initiatives on a firm's financial performance.

$$ROA_{i,t} = \beta_0 + \beta_1 ESGC_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (1)$$

Where $ROA_{i,t}$ is the dependent variable representing the firm's profitability of a company i , in time t . $ESGC_{i,t-1}$ represents the one-year lag variable of the ESG combined score of a company i , in time $t-1$. IND_i & $YEAR_t$ are the industry-fixed effects and time-fixed effects. $\varepsilon_{i,t}$ represents the error term where the models cannot explain parts of the observables.

6.3.2 Hypothesis 2

To test the second hypothesis, whether there is a difference in the ESG-CFP relationship between large firms and SMEs, the dichotomy of LNTA is used. All firms are divided into two categories, where large firms = 1 and SMEs = 0, using the mean of LNTA as a threshold. This division allows the separate testing of the effect of ESG performance on a firm's profitability. The following equations (1A and 1B) show the regression models for large firms and SMEs, which regress ROA against the one-year lagged ESG combined score and other specific firm characteristics.

Large firm = 1

$$ROA_{i,t} = \beta_0 + \beta_1 ESGC_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (1A)$$

SMEs = 0

$$ROA_{i,t} = \beta_0 + \beta_1 ESGC_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (1B)$$

Where $LNTA_{i,t}$ in the model (1A) and (1B) represent large firms and SMEs, respectively.

6.3.3 Hypothesis 3

It's worth noting that the disaggregated elements of ESG affect a firm's profitability differently, as indicated in Whelan et al. (2021). ESG elements might vary significantly depending on the specific industries or companies. Therefore, analyzing disaggregated ESG factors allows for a more detailed evaluation of which components contribute the most to the overall ESG performance in large firms and SMEs. The three following equations assist in testing the third hypothesis, which is the impact of the individual effects of each ESG component (E, S, and G) on all firms, large firms, and SMEs' financial performance.

Environment element:

$$ROA_{i,t} = \beta_0 + \beta_1 ENV_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (2)$$

Social element:

$$ROA_{i,t} = \beta_0 + \beta_1 SOC_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (3)$$

Governance element:

$$ROA_{i,t} = \beta_0 + \beta_1 GOV_{i,t-1} + \beta_2 LNTA_{i,t} + \beta_3 PPE_{i,t} + \beta_4 GROWTH_{i,t} + \beta_5 CASH_{i,t} + \beta_6 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (4)$$

Where $ENV_{i,t-1}$, $SOC_{i,t-1}$, and $GOV_{i,t-1}$ are the lagged variables of ESG individual elements of a company I in time t-1.

6.3.4 Non-linear analysis

Non-linearity analysis examines whether the relationship between ESG-CFP potentially presents a different pattern rather than a linear relationship (Brammer et al., 2008; Han et al., 2016; Naimy et al., 2021). The section adds the squared-lagged variable of the ESG combined score in the base model (1) for all firms, large firms, and SMEs, respectively.

$$ROA_{i,t} = \beta_0 + \beta_1 ESGC_{i,t-1} + \beta_2 ESGC^2_{i,t-1} + \beta_3 LNTA_{i,t} + \beta_4 PPE_{i,t} + \beta_5 GROWTH_{i,t} + \beta_6 CASH_{i,t} + \beta_7 LEVERAGE_{i,t} + IND_i + YEAR_t + \varepsilon_{i,t} \quad (1C)$$

Where $ESGC^2_{i,t-1}$ variable represents the square-lagged value of the ESG combined score and is used to test for non-linearity.

6.4 Diagnostic tests

This subsection proposes several necessary tests for panel data analysis. Stata and R software are used to perform these tests to enhance the results.

Multicollinearity

Multicollinearity occurs when there is a high level of linear correlation between the explanatory variables in the regression model. This correlation violates one of the OLS assumptions of no perfect collinearity and leads to misleading estimation results (Kim, 2019). A Pearson Correlation Matrix and a Variance Inflation Factor (VIF) analysis are conducted before the estimation procedure to test for multicollinearity in the data.

VIF analysis examines the degree of correlation among the independent variables. The VIF analysis was conducted by running linear regression on different sets of variables to estimate the inflation of a coefficient caused by a linear dependence on other predictors. A common rule of thumb for no serious multicollinearity is that the values of VIF must be less than 10 (Shrestha, 2020). No sign of multicollinearity was detected between different variables, as presented in [Appendix D](#), as all the VIF values are less than 2.

Pearson Correlation Matrix assesses the correlation between explanatory variables. If a high correlation exists between the explanatory variables, collinearity is more likely when the correlation coefficient approaches 0.8 (Shrestha, 2020). [Appendix E](#) presents the Pearson Correlation Matrix for the explanatory variables in the analysis. It's noteworthy that the one-year lagged variable and the variable itself have high correlations of 0.7; for example, $ESGC_{i,t-1}$ vs $ESGC_{i,t}$; and $ENV_{i,t-1}$ vs $ENV_{i,t}$. The aggregated ESG combined score and disaggregated E, S, and G are also highly correlated, correlating approaching 0.8, thus, these variables should not be included in one regression model.

Time-fixed effect:

Time-fixed effect controls for the unobserved time-invariant factors that can affect the ESG-CFP relationship but are constant over time. This effect also allows for estimating time trends, potentially capturing the systematic changes that occur over time. Usually, we suspect that there are time-specific events that affect all firms in the same way (Schmidheiny, 2023), such as the Covid lockdown. The equation with time-specific effect δ_t is expressed as below:

$$y_{i,t} = \alpha + X_{i,t}\beta_1 + Z_i\beta_2 + \delta_t + \mu_i + \epsilon_{i,t} \quad (7)$$

Where $y_{i,t}$ is the dependent variable; $X_{i,t}$ is the vector of the time-variant variables; Z_i is the vector of time-invariant variables; δ_t is the time-specific effect; μ_i is the individual-specific effects; $\epsilon_{i,t}$ is the idiosyncratic errors.

An F-test for the joint significance of the time-fixed effects is used as the post-estimation test for each fixed-effects model. The null hypothesis indicates that all time-fixed effects are jointly significantly equal to zero. A p-value smaller than 0.05 indicates a time-fixed effect in all models ([Appendix F](#)) except SME models. Therefore, the time-fixed effect should be included when modeling where the impact is significant.

Heteroskedasticity

Heteroskedasticity represents that the variance of the error term is not equal and depends on some exogenous variables. Specifically, heteroskedasticity means a systematic change in the spread of the residuals; therefore, this problem violates the OLS assumption of constant variance. When heteroskedasticity is present in the data, the individual-specific effects will not be homoscedastic. Estimators under heteroskedasticity are inefficient, and standard errors are biased (Williams, 2020).

The test results from the modified Wald test ([Appendix G](#)) support the presence of heteroskedasticity. If p-values are smaller than 0.05, we reject the null hypothesis of homoskedasticity, thus confirming that all models have heteroskedasticity. Robust standard errors are used to correct the presence of heteroskedasticity in the models as they relax on the assumption that the error terms are independent and identically distributed (Williams, 2020). Therefore, robust standard errors are more trustworthy when heteroskedasticity is present.

Autocorrelation

Autocorrelation, or serial correlation, refers to a phenomenon where a regression model's error terms or residuals are correlated across different periods. Autocorrelation violates one of the critical assumptions of a linear regression model, which is independent and identically distributed of the error terms. A linear panel model with autocorrelation leads to biases in the standard errors, thus affecting the reliability and validity of the analysis

(Drukker, 2003). Mathematically, autocorrelation occurs in the idiosyncratic errors when $E[\epsilon_{i,t}\epsilon_{i,s}] \neq 0$ for all $s \neq t$ in the linear one-way model:

$$y_{i,t} = \alpha + X_{i,t}\beta_1 + Z_i\beta_2 + \mu_i + \epsilon_{i,t} \quad (8)$$

Where $y_{i,t}$ is the dependent variable; $X_{i,t}$ is the vector of the time-variant variables; Z_i is the vector of time-invariant variables; μ_i is the individual-specific effects; $\epsilon_{i,t}$ is the idiosyncratic errors.

Autocorrelation was detected by implementing the Wooldridge test for serial correlation in panel data for each regression model in the analysis. The null hypothesis (H_0) indicates no first-order autocorrelation, hence rejecting H_0 means the model suffers from serial correlation. The test results in [Appendix H](#) strongly reject the H_0 of no autocorrelation in all firm and large firm models. In the case of SMEs, the model with disaggregated $ENV_{i,t-1}$ and $GOV_{i,t-1}$ show no sign of autocorrelation.

Since the models suffer from heteroskedasticity and autocorrelation, cluster robust standard errors, known as White estimators, are recommended to produce consistent estimates of standard errors (Baltagi, 2001; Wooldridge, 2002). White estimators allow for valid inference in the presence of heteroskedasticity and autocorrelation.

Cross-sectional dependence

Cross-sectional dependence is present where there is a correlation between different cross-sectional units, for example, firms, in the panel data. Specifically, observations across units within the same period are not independent. This phenomenon arises due to spillover effects, standard shocks, or unobserved heterogeneity that becomes part of the error term (De Hoyos & Sarafidis, 2006). Ignoring cross-sectional correlation may lead to inefficient estimates and biased inference. Mathematically, cross-sectional dependence occurs when

$\text{Cov}(\epsilon_{i,t}\epsilon_{j,t}) \neq 0$ for some t and some $t \neq j$ (Sarafidis & Wansbeek, 2012), considering the panel data model below:

$$y_{i,t} = \alpha + X_{i,t}\beta_1 + Z_i\beta_2 + \gamma_i + \epsilon_{i,t} \quad (9)$$

Where $y_{i,t}$ is the dependent variable; $X_{i,t}$ is the vector of the time-variant variables; Z_i is the vector of time-invariant variables; γ_i is the individual-specific time-invariant unobserved effects; $\epsilon_{i,t}$ is the errors that may be cross-sectional dependent.

There are several diagnostic tests for cross-sectional dependence, such as the LM test proposed by Breusch & Pagan (1980), Frees (1995), and Pesaran (2004) tests. Since the panel data in this analysis has a large N and small T, the latter three tests are preferred as LM statistics exhibit significant size distortions (De Hoyos & Sarafidis, 2006). Nevertheless, this analysis used Pesaran's CD test to detect cross-sectional dependence. The test results in [Appendix I](#) confirmed that cross-sectional correlation exists in all regression models except for all SME models.

Since cross-sectional dependence is present in the models, Driscoll-Kraay standard errors are applied to correct heteroskedasticity and autocorrelation instead of cluster robust standard errors alone (Hoechle, 2007). Driscoll & Kraay's (1998) estimators provide consistent heteroskedastic standard errors that are robust to spatial dependence without changing the signs of the coefficients.

7. Empirical Results

7.1 Descriptive statistics

The study's total sample consists of 1408 firm-year observations for 355 publicly listed UK firms from 2018 to 2022. Table 3-Table 5 shows the composition of the total firm sample by year and industry and the descriptive statistics for each variable across all firms. Table 3 shows the number of large firms and the number of SMEs in each sample year accordingly.

Table 3. Composition of firm size by sample year

Year	Large firms	SMEs	Sum
2018	169	186	355
2019	169	186	
2020	169	186	
2021	170	185	
2022	171	185	
Total observations	927	848	1775

Note: The table represents the composition of the sample year by firm size.

The size category of these firms has remained stable over the years, mainly due to using the mean of LNTA as the cutoff point and the majority rule classification. Consequently, firms consistently being categorized as large firms or SMEs over most of the years tend to maintain their categorization.

The name of the industry, the industry code, the number of observations in each sector, and its contribution to the total sample are displayed in Table 4. The name of the industry is classified according to the industry classification benchmark (ICB), offering a robust system for comparing organizations (FTSE Russell, 2024). ICB has a system that includes 11 industries. It's worth noting that the Financials, Energy, and Real Estate industries have already been removed from the analysis. The Industrials and Consumer Discretionary industries are ranked at the top based on having the highest frequencies of firms, which comprise 34.08% and 29.86% of the overall sample, respectively.

Table 4. Composition of each sample industry

Industry Code	ICB Industry	Observations	%
55	Basic Materials	140	7.89
40	Consumer Discretionary	530	29.86
45	Consumer Staples	155	8.73
20	Health Care	110	6.2
50	Industrials	605	34.08
10	Technology	155	8.73
15	Telecommunications	40	2.25
65	Utilities	40	2.25
Total		1775	100

Note: The table represents the ICB industry compositions of firms in the sample.

Table 5. Descriptive statistics for all firms in the sample

Explanatory variables	No.	Mean	Std	P25	Median	P75
$ESGC_{i,t-1}$	1493	48.10	18.56	34	49	61
$ENV_{i,t-1}$	1445	43.16	23.47	25	41	59
$SOC_{i,t-1}$	1493	49.92	22.71	32	49	68
$GOV_{i,t-1}$	1492	56.71	21.54	41	59	74
$ROA_{i,t}$	1765	6.69	8.35	2.34	6.68	11.74
$LNTA_{i,t}$	1774	13.51	1.77	12.13	13.17	14.68
$LEVERAGE_{i,t}$	1689	0.24	0.17	0.11	0.22	0.35
$PPE_{i,t}$	1761	0.72	0.85	0.16	0.40	0.89
$GROWTH_{i,t}$	1755	0.11	0.19	-0.006	0.07	0.18
$CASH_{i,t}$	1771	0.12	0.10	0.04	0.09	0.17

Note: The table represents descriptive statistics of all variables for all firms. $ESGC_{i,t-1}$ represents the one-year lagged variable of firms' aggregated ESG score. $ENV_{i,t-1}$, $SOC_{i,t-1}$, $GOV_{i,t-1}$ are the one-year lagged variables of firms' disaggregated ESG scores. Std is standard deviation statistics. P25 and P75 represent the 25th and 75th percentiles.

Variables descriptive statistics for all firms are provided in Table 5. These variables represent various aspects of ESG performance and other financial control variables. To control the effect of outliers in the sample, all variables are winsorized at the 5th and 95th percentiles (Ahmad et al., 2021; Gholami et al., 2022). The total number of observations

varies slightly across the ESG and financial variables, indicating missing ESG scores potentially caused by incomplete data reporting. The mean and the standard deviation of ESG aggregated and disaggregated scores support the representative nature of these scores, where these scores range from 0-100. The mean ROA indicates that, on average, all firms in the sample generate a positive EBIT on their assets, roughly 6.7% across five years. High standard deviations of ROA at 8.35% show considerable variability in the ROA across the sample, indicating that negative returns exist in these periods. It's also worth noting that the negative statistics in the GROWTH variable result from extreme volatility due to the periods used in the sample.

Table 6 below summarizes statistics for large firms (Panel A) and SMEs (Panel B).

Table 6. Panel A-statistics for large firms. Panel B-statistics for SMEs

Panel A						
Explanatory variables	No.	Mean	Std	P25	Median	P75
<i>ESGC_{i,t-1}</i>	803	56.67	16.05	46	57	68
<i>ENV_{i,t-1}</i>	797	52.15	22.46	36	52	71
<i>SOC_{i,t-1}</i>	803	60.59	20.33	45	62	76
<i>GOV_{i,t-1}</i>	803	65.38	18.21	54	68	80
<i>ROA_{i,t}</i>	847	6.42	7.26	2.56	6.30	10.80
<i>LNTA_{i,t}</i>	847	15.00	1.30	13.98	14.77	15.79
<i>LEVERAGE_{i,t}</i>	832	0.29	0.16	0.18	0.28	0.40
<i>PPE_{i,t}</i>	846	0.83	0.92	0.22	0.52	0.95
<i>GROWTH_{i,t}</i>	842	0.09	0.17	-0.01	0.05	0.15
<i>CASH_{i,t}</i>	846	0.10	0.08	0.04	0.08	0.13
Panel B						
Explanatory variables	No.	Mean	Std	P25	Median	P75
<i>ESGC_{i,t-1}</i>	690	38.12	16.17	26	37	49
<i>ENV_{i,t-1}</i>	648	32.10	19.65	16	30	45
<i>SOC_{i,t-1}</i>	690	37.50	18.68	24	36	49
<i>GOV_{i,t-1}</i>	689	46.60	20.72	31	47	62
<i>ROA_{i,t}</i>	918	6.94	9.24	1.99	7.13	12.56
<i>LNTA_{i,t}</i>	927	12.15	0.76	11.65	12.17	12.77
<i>LEVERAGE_{i,t}</i>	857	0.20	0.16	0.06	0.17	0.29
<i>PPE_{i,t}</i>	915	0.62	0.78	0.12	0.29	0.77
<i>GROWTH_{i,t}</i>	913	0.12	0.20	0.0008	0.09	0.22
<i>CASH_{i,t}</i>	925	0.14	0.11	0.05	0.11	0.21

Note: The table presents the descriptive statistics of variables for large firms and SMEs in the sample.

On average, ESG combined score ranges from 38.12 for SMEs to 56.67 for large firms. Variations range from 26 for SMEs to 46 for large firms at the 25th percentile, while they range from 49 to 68 at the 75th percentile for SMEs and large firms, respectively. The

established statistics provided in Table 6 show sufficient variation in the ESG combined score to examine the effect of a firm's size on the relationship between ESG-CFP. The average ROA does not differ significantly across a firm's size, but ROA's variability shows notable distinction. While large firms' ROA has a standard deviation of 7.26%, SMEs demonstrate much higher volatility due to more significant variations in their standard deviation, at 9.24%. Interestingly, the mean value of GROWTH and CASH for SMEs are higher than large firms, indicating potentially higher growth rates and cash reserves among SMEs in the sample.

7.2 Results

7.2.1 ESG Performance and Profitability

ROA, or BEP ratio, is used as an indicator of firms' profitability to test the first baseline estimation model. The time effect is included due to its significance in the model, which was tested by an F-test joint significance in [Appendix F](#). The results of the first estimation model are presented in Table [Z](#).

Table 7. Baseline regression – aggregated ESG on ROA

Variables	ROA
$ESGC_{i,t-1}$	-0.043*** (0.0044)
$LNTA_{i,t}$	-1.23 (0.99)
$LEVERAGE_{i,t}$	-10.2*** (0.50)
$PPE_{i,t}$	-3.33** (0.42)
$GROWTH_{i,t}$	7.07** (1.36)
$CASH_{i,t}$	6.51 (3.38)
Constant	29.9 (13.5)
Time effects	Yes
Observations	1408
R^2	0.7366
Adjusted R^2	0.6453
F-test (fixed effects)	7.435***

Note: This table shows the main regression result of the combined ESG score on firms' profitability. The model was estimated using Driscoll-Kraay standard errors to account for cross-sectional dependence. Standard errors in parentheses. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Firms' ESG performance coefficient of -0.043 is statistically significant at 1% level significance (p-value = 0.007). On a scale from 0 to 100 of the ESG score, a one-unit change in the ESG score would represent a change of one point. Simply put, the results show that a one-point increase in the firm's ESG performance in the previous year leads to a 0.043 percentage point decrease in ROA in the current year. The result contradicts the study's expectation and other literature findings (Ahmad et al., 2021; Dellaportas et al., 2012; Gholami et al., 2022), where they found a positive correlation between ESG-CFP. Therefore, it does not support the first hypothesis that higher ESG performance drives higher profitability. However, some researchers found similar findings, indicating that firms' ESG

score negatively affects their corresponding CFP (Brammer et al., 2008; Nirino et al., 2021; Seifert et al., 2003). The findings align with certain aspects of “agency theory,” emphasizing that when companies prioritize ESG investing, it may divert financial resources away from activities that directly benefit and maximize shareholder value. Therefore, managers who are considered agents and act on behalf of the principal, those shareholders, could potentially decrease financial performance.

However, the findings for other variables align with the current findings in the literature, where their coefficients follow the same direction. Significantly negative correlations between ROA, LEVERAGE, and PPE reflect higher levels of financial risks and imply increased volatility in the firm’s profitability. These negative associations can be explained by high debt levels and intense capital investments (Bauwhede, 2009; Musah & Kong, 2019). In contrast, a positive correlation is found for ROA, GROWTH, and CASH. These results align with King and Santor (2008), emphasizing cash reserves as strategic resources for a firm’s flexibility and investment opportunities. However, as measured by the log of total assets (LNTA), firm size has a negative and weak but insignificant correlation with ROA. This correlation is consistent with Bauwhede’s (2009) findings, which suggested that profitability decreases as firms grow. One possible explanation is the theory of diminishing returns to scale, where the production process becomes inefficient and expensive as firm size increases.

R-squared (R^2) and adjusted R-squared (adjusted R^2) values represent the model’s overall fit. While R^2 specifies how much variance in ROA can be explained by the model-independent variables, adjusted R^2 also account for the number of predictors in the model. For example, approximately 64.53% of the variance in ROA is explained after adjusting for the number of predictors. However, it’s worth noting that fixed-effect model estimating in Stata provides incorrect R^2 and adjusted R^2 values (Park, 2011). Hence, the research

employed Linear Regression with a large dummy variable set (using the *areg* command in Stata) to provide a more appropriate and accurate calculation of R^2 and adjusted R^2 for all models, as recommended by Park (2011).

7.2.2 Firm Size Analysis

Table 8 below presents the results of the firm's size analysis between large firms (Model 1A) and SMEs (Model 1B) in testing H2 as hypothesized.

Table 8. Large vs SMEs baseline comparison analysis – aggregated ESG on ROA

Variables	ROA (1A)	ROA (1B)
$ESGC_{i,t-1}$	-0.025* (0.0073)	-0.070* (0.029)
$LNTA_{i,t}$	-0.10 (1.37)	-1.91 (1.31)
$LEVERAGE_{i,t}$	-10.6** (1.25)	-9.57* (4.55)
$PPE_{i,t}$	-2.54* (0.90)	-4.90*** (1.44)
$GROWTH_{i,t}$	6.40* (1.55)	7.75*** (1.88)
$CASH_{i,t}$	-2.43 (3.87)	12.9 (6.62)
Constant	14.8 (20.7)	36.4* (15.8)
Time effects	Yes	No
Observations	782	626
R^2	0.7346	0.7442
Adjusted R^2	0.6551	0.6300
F-test (fixed effects)	7.921***	6.317***

Note: This table shows the main regression result of the combined ESG score on firms' profitability. The first model (1A) reports Driscoll-Kraay standard error estimation for large firms, and the second model (1B) reports the fixed effects with robust and cluster standard errors for SMEs. Standard errors in parentheses. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The findings from model 1A and model 1B highly support the third hypothesis, which states a distinct difference between large and SMEs regarding their ESG-CFP relationship. Both models show significantly lagged ESG combined score variables, emphasizing a negative correlation between ESG performance and a firm's profitability. Both adjusted R-squared values exceed over 60%, implying relatively substantial explanatory power of all the variables in the models. However, the magnitude of the effect appears to be more significant for SMEs despite the two models showing the same statistical significance at 5%. In model 1B, SMEs exhibit a more substantial effect of ESG performance on ROA, with a coefficient

of -0.07 (p-value = 0.015) compared to -0.025 (p-value = 0.025) for large firms. Model 1B indicates that a point unit increase in the ESG performance for SMEs in the last year is associated with a 0.07% decrease in profitability compared to a slight decline of 0.025 % in large firms, as evidenced by Model 1A. This negative correlation again contradicts the positive findings. It suggests that an improvement in ESG performance might have a delayed negative impact on profitability, possibly due to cost implications, market dynamics, and significant events such as the COVID-19 pandemic.

The negative correlation between ROA, LEVERAGE, and PPE in both models, displayed in Table 8, is consistent with findings from Table 7. The negative coefficient of LEVERAGE suggests that higher debt levels negatively affect firms' performance in terms of profitability (Bauwhede, 2009; Yazdanfar & Öhman, 2015). Larger firms tend to take on more debt, which negatively affects their ROA, as evidenced by the LEVERAGE coefficient of -10.6 (p-value = 0.001) compared to -9.57 (p-value = 0.037) for SMEs. SMEs tend to finance their business with equity to a high degree (Yazdanfar & Öhman, 2015). A higher investment in PPE also potentially lowers ROA, which puts more constraints on SMEs than large firms, as evidenced by the PPE coefficient of -4.90 (p-value = 0.001) compared to -2.54 (p-value = 0.048) respectively. When these assets are not utilized effectively or become obsolete, they may not generate enough returns to cover their costs, thus lowering ROA.

Conversely, GROWTH exhibits a higher impact on profitability in SMEs than large firms. The growth coefficient for SMEs has a relatively higher impact on ROA, which is 7.75 (p-value = 0.000), and is statistically significant at a 1% level, while the growth coefficient for large firms is 6.40 (p-value = 0.015), and statistically significant at 5% level. This positive relationship between growth and ROA and the magnitude of sales growth in SMEs' cases is also supported by Mijić et al. (2018).

Notably, the coefficient of the CASH ratio shows contradicting signs between models, which are negative for large firms and positive for SMEs. A negative coefficient in large firms may suggest inefficient utilization of resources, as cash generates lower returns than other investments. In contrast, SMEs tend to have more cash holdings than their larger counterparts. Higher cash reserves can sustain firms' viability and financial performance, especially during an economic downturn (Dimitropoulos et al., 2020). However, CASH coefficients are not statistically significant in both models.

7.2.3 Disaggregated Analysis of ESG

Table 9-Table 11 present additional analysis required to assess which of the three disaggregated elements of ESG affect all firms, large firms, and SMEs' financial performance.

Table 9. Disaggregated ESG analysis – All firms

Variables	ROA			
	(1)	(2)	(3)	(4)
$ESGC_{i,t-1}$	-0.043*** (0.0044)			
$ENV_{i,t-1}$		-0.018** (0.0034)		
$SOC_{i,t-1}$			-0.031* (0.0067)	
$GOV_{i,t-1}$				-0.023*** (0.0018)
$LNTA_{i,t}$	-1.23 (0.99)	-1.17 (1.10)	-1.22 (1.04)	-1.31 (1.02)
$LEVERAGE_{i,t}$	-10.2*** (0.50)	-9.43*** (0.58)	-10.1*** (0.48)	-9.77*** (0.56)
$PPE_{i,t}$	-3.33** (0.42)	-3.44** (0.42)	-3.37** (0.43)	-3.34** (0.43)
$GROWTH_{i,t}$	7.07** (1.36)	7.22** (1.27)	7.01** (1.39)	7.34** (1.27)
$CASH_{i,t}$	6.51 (3.38)	4.43 (3.79)	6.25 (3.40)	6.02 (3.48)
Constant	29.9 (13.5)	27.9 (15.0)	29.1 (14.3)	30.2 (13.8)
Time effect	Yes	Yes	Yes	Yes
Observations	1408	1368	1408	1407
R^2	0.7366	0.7426	0.7360	0.7421
Adjusted R^2	0.6453	0.6510	0.6445	0.6527
F-test (fixed effects)	7.435***	7.442***	7.434***	7.689***

Note: This table shows the main regression result of aggregated and disaggregated ESG scores on all firms' profitability. All models report Driscoll-Kraay standard error estimation for the ESG combined score and environmental, social, and governance scores. Standard errors in parentheses. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 10. Disaggregated ESG analysis – Large firms

Variables	ROA			
	(1)	(2)	(3)	(4)
$ESGC_{i,t-1}$	-0.025** (0.0073)			
$ENV_{i,t-1}$		0.00056 (0.0053)		
$SOC_{i,t-1}$			-0.020 (0.013)	
$GOV_{i,t-1}$				-0.013** (0.0047)
$LNTA_{i,t}$	-0.10 (1.37)	-0.052 (1.38)	-0.14 (1.42)	-0.074 (1.39)
$LEVERAGE_{i,t}$	-10.6*** (1.25)	-10.6**** (1.18)	-10.5*** (1.36)	-10.5*** (1.28)
$PPE_{i,t}$	-2.54** (0.90)	-2.60** (0.90)	-2.59 (0.96)	-2.56 (0.94)
$GROWTH_{i,t}$	6.40** (1.55)	6.35** (1.52)	6.35** (1.54)	6.35** (1.53)
$CASH_{i,t}$	-2.43 (3.87)	-3.35 (4.06)	-2.76 (3.89)	-2.55 (3.98)
Constant	14.8 (20.7)	12.8 (20.7)	15.0 (21.9)	13.8 (21.0)
Time effect	Yes	Yes	Yes	Yes
Observations	782	777	782	782
R^2	0.7346	0.7341	0.7344	0.7339
Adjusted R^2	0.6551	0.6539	0.6549	0.6542
F-test (fixed effects)	7.921***	7.829***	8.010***	8.084***

Note: This table shows the main regression result of aggregated and disaggregated ESG scores on large firms' profitability. Models (1), (2), (3), and (4) report Driscoll-Kraay standard error estimation for ESG combined score, and environmental, social, and governance score, respectively. Standard errors in parentheses. Superscripts indicate significance level * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, **** $p < 0.001$.

Table 11. Disaggregated ESG analysis – SME firms

Variables	ROA			
	(1)	(2)	(3)	(4)
$ESGC_{i,t-1}$	-0.070** (0.029)			
$ENV_{i,t-1}$		-0.038 (0.023)		
$SOC_{i,t-1}$			-0.040* (0.021)	
$GOV_{i,t-1}$				-0.034* (0.020)
$LNTA_{i,t}$	-1.91 (1.31)	-2.02 (1.37)	-1.76 (1.34)	-2.16 (1.24)
$LEVERAGE_{i,t}$	-9.57** (4.55)	-8.13 (4.67)	-9.51** (4.63)	-8.87** (4.44)
$PPE_{i,t}$	-4.90**** (1.44)	-4.97**** (1.46)	-4.86**** (1.42)	-4.86**** (1.38)
$GROWTH_{i,t}$	7.75**** (1.88)	8.29**** (1.80)	7.65**** (1.88)	8.46**** (1.72)
$CASH_{i,t}$	12.9 (6.62)	10.6 (6.79)	13.1** (6.61)	12.6 (6.59)
Constant	36.4** (15.8)	36.2** (16.5)	32.9** (16.1)	37.7** (14.9)
Time effect	No	No	No	No
Observations	626	591	626	625
R^2	0.7442	0.7579	0.7422	0.7557
Adjusted R^2	0.6300	0.6399	0.6271	0.6462
F-test (fixed effects)	6.317***	6.335*	6.244***	6.694***

Note: This table shows the main regression result of aggregated and disaggregated ESG scores on SME firms' profitability. Models (1), (2), (3), and (4) report fixed-effects models with robust and cluster standard error for ESG combined score, and environmental, social, and governance scores, respectively. Standard errors in parentheses. Significance level * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, **** $p < 0.001$.

All firms

The result from Table 9 reveals a robust association between a firm's disaggregated ESG performance and its profitability, with each coefficient element of ESG being significant at varying levels, 1%, 5%, and 0.1%, respectively. These findings align with prior literature on stakeholder theory, which emphasizes the crucial role of sustainability in a firm's profitability. However, negative coefficients of these disaggregated scores, -0.018, -0.031, and -0.023, respectively, signify that ROA would decrease to a percentage point for every one-point unit increase in each element. For instance, a one-point unit increase in the environmental and social score in the previous year led to a decrease of 0.018% or 0.031% in firms' ROA. These significant adverse effects suggest that spending on ESG elements decreases the current profitability of the firms.

These negative correlations of ESG elements are inconsistent with studies that found a significantly positive relationship between corporate environmental, social, governance performance, and financial performance (Affes & Jarboui, 2023; Moneva & Ortas, 2010; Orlitzky et al., 2003). Nevertheless, Fisher-Vanden & Thorburn's (2011) findings support the negative correlation between CEP and ROA. Their results indicated that ecological investment is the firm's internal financial burden. Environmental investment takes a long time to gain benefits, thus deteriorating economic performance. Environmental investment, however, is likely to positively influence the market performance indicator, such as Tobin's Q, as stock prices reflect a firm's environmental initiative.

CSP was also evidenced to negatively influence CFP from Brammer et al.'s (2008) and Lankoski's (2000) findings. The latter research confirmed that the CSP-CFP relationship is complex due to the size of sustainability costs. Notably, CGP exhibits the most substantial impact on ROA but is negative, confirming the second hypothesis. The result implies that corporate governance is the primary driver of corporate sustainability, consistent with Friede

et al. (2015). The result also strongly supports agency theory's implication, where effective corporate governance could mitigate agency problems caused by conflicts of interest (Chen et al., 2011).

The firm's size (LNTA) shows the same negative sign and magnitude across model 2, model 3, and model 4. Debt ratio (LEVERAGE) and property, plant, and equipment (PPE) show consistent and significantly negative associations with ROA in all three ESG components models. A firm's revenue growth (GROWTH) and liquidity (CASH) positively correlate with ROA. Their correlations with ROA are consistent across the three disaggregated ESG models, as evidenced by their same coefficient signs.

Large firms and SMEs

The comparison of the three elements of ESG performance between large firms (Table 10) and SMEs (Table 11) reveals distinct patterns in their relationship with ROA, which are different from the findings found for all firms. The lagged variable of governance score continues to show a negative association with ROA for both firms, consistent with the results for all firms. However, the effect is more substantial for SMEs, which is -0.034 (p-value = 0.096) compared to -0.013 (p-value = 0.05). The effect implies a more pronounced importance of corporate governance in SMEs' profitability. In contrast, the environmental score is positively associated with ROA for large firms but negative for SMEs, despite neither of the two models showing its significance.

Similarly, although social score coefficients negatively correlate with ROA for both, the coefficient is only significant in the case of SMEs. For every one-point unit increase in the social score in the previous year, SMEs' profitability is expected to decrease by 0.04% (p-value = 0.064). These findings align with Arend's (2014) findings, where he emphasized the critical influence of social and governance performance on SMEs' profitability. At the same

time, some tradeoffs still exist between activities aimed at financial performance and improving sustainability performance.

In the disaggregated ESG analysis for large firms and SMEs, other financial control variables continue to show consistent results. LEVERAGE and PPE are significantly negatively associated with firms' profitability in nearly three ESG components models, except for insignificant coefficients of PPE in model 3 and model 4 for large firms. GROWTH consistently shows a favorable relationship with ROA across large firms and SME analyses. Finally, no significant associations are found between LNTA, CASH, and the firms' profitability for both groups.

7.2.4 Non-linearity analysis

Given the significant negative association between ESG performance and a firm's profitability, evaluating whether this association presents a different pattern is necessary. Non-linearity analysis suggests that the impact of ESG initiatives on profitability doesn't follow a simple linear trend but instead fluctuates in a more complex direction. Table 12 below presents the regression results to assess the non-linear relationship between ESG performance and CFP by including the square-lagged variable of the ESG combined score.

Table 12. Non-linear relationship of ESG performance and CFP

Variables	ROA		
	ALL FIRMS	LARGE FIRMS	SMEs
$ESGC_{i,t-1}$	-0.12*** (0.016)	-0.049 (0.047)	-0.16*** (0.027)
$ESGC^2_{i,t-1}$	0.00083*** (0.00011)	0.00022 (0.00036)	0.0013** (0.00033)
$LNTA_{i,t}$	-1.24 (0.95)	-0.11 (1.36)	-1.60** (0.54)
$LEVERAGE_{i,t}$	-10.3**** (0.47)	-10.5*** (1.27)	-11.2*** (1.52)
$PPE_{i,t}$	-3.28*** (0.43)	-2.55** (0.91)	-4.51*** (0.72)
$GROWTH_{i,t}$	7.12*** (1.33)	6.40** (1.56)	9.03*** (1.21)
$CASH_{i,t}$	6.24*** (3.40)	-2.51 (3.81)	11.7** (3.98)
Constant	31.8* (13.23)	15.5 (21.44)	33.1*** (5.88)
Time effect	Yes	Yes	No
Observations	1408	782	626
R^2	0.7373	0.7347	0.7458
Adjusted R^2	0.6460	0.6546	0.6314

Note: This table shows the additional regression result of non-linear analysis between ESG performance and CFP for all firms, large firms, and SMEs, respectively. All models report Driscoll-Kraay standard error estimation. Standard errors in parentheses. Significance level * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, **** $p < 0.001$

The result indicates that there is a significant non-linear relationship between ESG performance and a firm's profitability across different firm sizes, which was confirmed by the coefficient values of the square-lagged variables ($ESGC^2_{i,t-1}$). The additional analysis reveals a significant non-linear relationship for all firms and SMEs, p-value = 0.002 and p-value = 0.017, respectively, suggesting that the impact of ESG factors on profitability varies nonlinearly according to the changes in ESG scores. Interestingly, this non-linear relationship

is not observed in large firms, indicating that the effects of ESG performance on financial outcomes may be more linear or less pronounced in these firms. Overall, the results imply that the impact of ESG factors on profitability varies depending on the firm's scale.

In terms of the coefficient sign of $ESGC^2_{i,t-1}$, positive coefficient across models suggests a convex (U curve) relationship. This means that the effect of ESG performance on financial performance initially must have a diminishing return or even negative to a certain point for it eventually to slowly increase or sometimes reverse as the squared ESG score rises. This finding aligns with McWilliams & Siegel's (2001) and Han et al.'s (2016) findings. Here, they indicated that the benefits of implementing ESG initiatives might not immediately come into effect and that they might take time to be cultivated. ESG investment is likely associated with significant capital investment, leading to an economy of scale. Given this argument, ESG investment must increase to a certain point to generate financial returns. Additionally, the signs of other financial control variables are consistent across models compared to Table 9-Table 11, indicating a reliable, robust analysis using different model specifications. These findings contribute to understanding the complex correlation between ESG performance and CFP. It highlights the importance of patience and long-term commitment in achieving sustainability goals and the need to consider firm size to tailor ESG strategies and policies.

8. Discussion & Conclusion

The study investigated whether investing in ESG activities enhances a firm's financial performance and whether this association is distinct between large firms and SMEs in the 355 listed UK firms from 2018 to 2022. By expanding an understanding of agency, institutional, and resource-based view theories, this study underscores the importance of firms' ESG performance in improving their sustainable behavior nationally and internationally. As noted earlier, agency theory highlights the importance of corporate governance in guiding business ethics. Recognizing and integrating ESG principles into a company's strategy can be seen as a unique and competitive resource from a resource-based view perspective. Concentrated ownership structure in SMEs could also be an advantage for encouraging ESG investing due to direct management compared to large firms. Hence, the institutional environment must create opportunities to facilitate the integration of ESG in the case of SMEs. Poor ESG efforts could threaten the firm's survival, whereas strong ESG performance is more likely to promote the company's reputation and accountability in the long run. Firm managers and national regulators can both benefit from the study's insights. SME managers can potentially change their focus towards long-term sustainable investment. In contrast, UK regulators can recognize most SMEs' financial constraints and impose measures to facilitate their sustainable journey.

The study focused on the overall performance of ESG scores. It expanded the analysis by better understanding the association between the individual element of ESG scores and the firm's profitability. These elements are environmental, social, and governance scores. The panel data regression model, specifically the fixed-effect model, mitigates the effects of time and temporal lags, controls for the specific unobserved-firm characteristics, and provides a more dynamic ESG-CFP relationship. The analysis provides a robust estimation using

Driscoll-Kraay estimators to control heteroskedasticity in the presence of cross-sectional dependence.

The first hypothesis did not support a positive association between a firm's aggregated ESG performance and its profitability, as found by some other studies (Ahmad et al., 2021; Gholami et al., 2022). While this study did not find evidence to support the expected positive relationship, it is also worth noting that its findings diverged from the initial expectation. This distinction highlights the need to consider the study context, especially the volatility period of the study and the nature of the firm's ESG data. Firstly, the study used the time frame from 2018 to 2022, which was highly volatile considering the Covid-19 pandemic and post-Covid economic crisis. In 2020, the first national lockdown in the UK severely interrupted its economic outlook due to a high unemployment rate and low household spending. As a result of damage related to the COVID-19 pandemic, businesses were forced to lay off employees or even exit the market (Allas et al., 2020). This posed a significant threat to the firm's financial performance and may restrict its sustainability activities. Secondly, as mentioned earlier, the lack of ESG data on these UK-listed firms significantly influenced the study's outcomes. In the initial sample, most firms did not report ESG disclosure until 2018, while some started to disclose their ESG scores in 2020. For this reason, the analysis narrowed its focus to the periods when most firms disclosed their ESG data. Therefore, the uneven ESG disclosure between firms posed challenges in assessing the long-term effect of ESG performance on profitability.

However, existing studies help explain the negative association between ESG performance and CFP. Hamdi et al. (2022) found that the relationship between ESG score and financial performance is significantly negative under high economic and political uncertainty. They explained that high leverage reduces governance practice spending during these difficult times. In other words, with market turmoil pre- and post-COVID pandemic, the

benefits of investing in ESG have been blunt because cash flow towards maintaining business operations is more critical than costly ESG investment (Zhang, 2022). Given the study's relative size (n=1408), the results are challenged through traditional stakeholders' theory and institutional theory, which suggest financial outcomes would improve accordingly with the improvement of ESG performance motivated by various stakeholders.

The second hypothesis on the association between ESG and CFP is confirmed through the analysis. The relationship between ESG-CFP is significantly distinct for large firms and SMEs, which is consistent with the findings of Gholami et al. (2022). While negative correlations between ESG-CFP were found, the non-linear relationship is more pronounced in the case of SMEs. The study contributes to the SMEs' existing body of literature by highlighting that the correlation between ESG performance and CFP presents a significant convex relationship in SMEs. This U-shape relationship implies that ESG investing is initially costly, and the investment must increase beyond a certain level to positively impact the firm's profitability. This significant non-linear relationship may also suggest a cost that exceeds SMEs' available resources, as Bianchi et al. (2015) found. In other words, the increased expense of collecting and disclosing sustainability information exceeds the incremental increase in profitability. Simply put, SMEs lack the same level of collecting and reporting ESG data as their large counterparts and stakeholders' engagement and feedback to implement and improve sustainability. With different characteristics, such as a small customer base and limited access to the capital market, SMEs may not practice the same level of ESG disclosure because it does not provide the same level of impact on their profitability compared to larger firms.

While the results did not confirm a win-win scenario, they shed light on various factors influencing the relationship, including the role of corporate governance and national regulators. The results for testing the third hypothesis confirm that corporate governance is

the primary force of ESG practices regardless of firm size, consistent with prior research (Affes & Jarboui, 2023; Kyere & Ausloos, 2020). It also supports the motivation explanation of agency and stakeholder theories, where effective corporate governance mitigates information asymmetry and reduces agency problems. By demonstrating the importance of corporate governance, the study contributes to understanding the dynamic motivation of fostering business ESG commitment and the interaction of different ESG elements on CFP. It highlights that organizations need a robust governance framework to foster ESG commitments.

8.1 Implication for practice

The study has several implications for firms' managers, market participants, and institutional regulators. This work presents the economic favor of promoting and enhancing ESG initiatives regardless of financial outcome. Given the significant benefits ESG investing plays in building a positive reputation and trust among different stakeholders, this also allows firms to participate in the country's sustainability journey. The association between ESG performance and a firm's financial performance seems to be twofold, contributing to a moral cycle. Investing in ESG activities improves a firm's financial performance; on the other hand, the firm's performance also reflects the level of sustainable commitments. Therefore, ESG investing should be promoted cautiously to ensure that the costs of being socially responsible do not outweigh the possible gains to avoid any outcomes from over-investing. On the other hand, a shift in the manager's mindset towards better sustainability development is also necessary for a long-term commitment. Sustainability development requires ongoing dedication; thus, managers should integrate sustainability into long-term corporate strategies as it can lead to environmental and financial benefits.

Regulators in the UK should continue to promote and facilitate firms to be socially responsible by creating a supportive institutional environment. Additionally, according to the

significant non-linear relationship of ESG-CFP in SMEs, a more tailored ESG strategy and disclosure standard may result in more future adoption of ESG initiatives. Given SMEs' substantial financial burden when investing in sustainability, regulators should carefully design a tailored ESG strategy to provide necessary resources and tools such as financial support and reporting guidance.

8.2 Implication for future research

As a relatively volatile period was utilized in the analysis, future research is recommended to continue to monitor and investigate the relationship between ESG-CFP in the UK over a more extended period with a larger sample. By considering a longer timeframe, more ESG data for firms becomes available and abundant. More data enhances a more pronounced finding of the observed ESG-CFP relationship without being affected by market disturbance, such as the COVID-19 pandemic. Moreover, the ESG-CFP relationship can also be influenced by the moderating role of other variables, such as an ESG committee, economic volatility, and environmental innovation through a firm's research and development. Future research with a moderating variable is highly recommended because it captures the ESG-CFP relationship alone and further enhances and reflects a more in-depth association. It would also be interesting to assess this relationship using different profitability measurements such as Tobin's Q, market value, or earning per share.

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Appendix

Appendix A. F-test for fixed effects

	All firms		Large firms		SMEs	
	F-test	P-value	F-test	P-value	F-test	P-value
Model 1	7.19	0.00 ***	7.72	0.00 ***	6.10	0.00 ***
Model 2	7.17	0.00 ***	7.60	0.00 ***	6.03	0.00 ***
Model 3	7.19	0.00 ***	7.87	0.00 ***	6.06	0.00 ***
Model 4	7.46	0.00 ***	7.67	0.00 ***	6.53	0.00 ***

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The table shows that the fixed-effects model is preferred over OLS due to the significance of improved goodness-of-fit using the fixed-effects model ($p < 0.01$).

Appendix B. Breusch and Pagan LM test for random effects

	All firms		Large firms		SMEs	
	Chi2	P-value	Chi2	P-value	Chi2	P-value
Model 1	646.92	0.00 ***	403.66	0.00 ***	249.42	0.00 ***
Model 2	647.54	0.00 ***	420.67	0.00 ***	241.61	0.00 ***
Model 3	664.70	0.00 ***	428.16	0.00 ***	252.00	0.00 ***
Model 4	694.98	0.00 ***	446.71	0.00 ***	268.40	0.00 ***

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The table shows that the random effects model is preferred over OLS due to the significant presence of random effects in all models ($p < 0.01$).

Appendix C. Durbin-Wu-Hausman test

	All firms		Large firms		SMEs	
	Chi2	P-value	Chi2	P-value	Chi2	P-value
Model 1	50.4	0.00 ***	37.21	0.00 ***	33.07	0.00 ***
Model 2	31.03	0.00 ***	23.63	0.01 ***	33.88	0.00 ***
Model 3	34.82	0.00 ***	25.47	0.00 ***	31.67	0.00 ***
Model 4	18.66	0.04 *	16.98	0.07	31.29	0.00 ***

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The null hypothesis indicates that the coefficient difference between the fixed and random effect models is not systematic. Rejecting the null hypothesis means there is a significant difference; hence, the fixed-effects model should be used over the random-effects model. Test results of all models confirm the use of the fixed-effects model throughout the analysis.

Appendix D. Variance Inflation Factor (VIF) analysis

Variables	VIF	1/VIF	VIF	1/VIF	VIF	1/VIF	VIF	1/VIF
$ESGC_{i,t-1}$	1.40	0.71						
$ENV_{i,t-1}$			1.42	0.71				
$SOC_{i,t-1}$					1.53	0.65		
$GOV_{i,t-1}$							1.32	0.76
$LNTA_{i,t}$	1.56	0.64	1.59	0.63	1.68	0.60	1.49	0.67
$LEVERAGE_{i,t}$	1.24	0.80	1.23	0.81	1.24	0.80	1.24	0.80
$PPE_{i,t}$	1.17	0.86	1.16	0.86	1.16	0.86	1.17	0.86
$GROWTH_{i,t}$	1.03	0.97	1.02	0.98	1.03	0.97	1.03	0.97
$CASH_{i,t}$	1.13	0.89	1.12	0.89	1.13	0.89	1.13	0.88

Note: $ESGC_{i,t-1}$: one-year lagged variable of ESG combined score; $ENV_{i,t-1}$: one-year lagged variable of Environmental Score; $GOV_{i,t-1}$: one-year lagged variable of Governmental Score; $SOC_{i,t-1}$: one-year lagged variable of Social Score. The table presents the VIF analysis using a linear regression model. VIF analysis is used to estimate the inflation of a coefficient due to linear dependence on other predictors. A VIF value higher than 10 is a severe multicollinearity that needs further investigation. All VIF values presented in the table are below 2.

Appendix E. Pearson correlation matrix of variables

	$ROA_{i,t}$	$ESGC_{i,t-1}$	$ESGC_{i,t}$	$ENV_{i,t-1}$	$ENV_{i,t}$	$GOV_{i,t-1}$	$GOV_{i,t}$	$SOC_{i,t-1}$	$SOC_{i,t}$	$LNTA_{i,t}$	$LEVERAGE_{i,t}$	$PPE_{i,t}$	$GROWTH_{i,t}$	$CASH_{i,t}$
$ROA_{i,t}$	1													
$ESGC_{i,t-1}$	0.05	1												
$ESGC_{i,t}$	0.06	0.79	1											
$ENV_{i,t-1}$	0.07	0.79	0.66	1										
$ENV_{i,t}$	0.07	0.66	0.76	0.83	1									
$GOV_{i,t-1}$	0.02	0.75	0.59	0.51	0.43	1								
$GOV_{i,t}$	0.02	0.60	0.71	0.43	0.47	0.74	1							
$SOC_{i,t-1}$	0.02	0.86	0.71	0.74	0.62	0.60	0.50	1						
$SOC_{i,t}$	0.04	0.74	0.83	0.65	0.72	0.53	0.55	0.84	1					
$LNTA_{i,t}$	-0.05	0.56	0.56	0.60	0.64	0.51	0.50	0.61	0.68	1				
$LEVERAGE_{i,t}$	-0.19	0.19	0.21	0.18	0.22	0.10	0.12	0.21	0.23	0.31	1			
$PPE_{i,t}$	-0.17	0.03	0.05	0.05	0.09	-0.02	-0.02	0.09	0.12	0.18	0.34	1		
$GROWTH_{i,t}$	0.18	-0.13	-0.17	-0.09	-0.08	-0.12	-0.16	-0.15	-0.18	-0.13	-0.10	-0.09	1	
$CASH_{i,t}$	0.06	-0.14	-0.14	-0.11	-0.11	-0.15	-0.15	-0.16	-0.18	-0.26	-0.23	-0.21	0.08	1

Note: $ESGC_{i,t-1}$: one-year lagged variable of ESG combined score; $ESGC_{i,t}$: ESG combined score; $ENV_{i,t-1}$: one-year lagged variable of Environmental Score; $ENV_{i,t}$: Environmental Score; $GOV_{i,t-1}$: one-year lagged variable of Governmental Score; $GOV_{i,t}$: Governmental Score; $SOC_{i,t-1}$: one-year lagged variable of Social Score; $SOC_{i,t}$: Social score. This table provides a correlation matrix of all variables. All variables show correlation of less than 0.6, with some exceptions. The lagged ESG variables and ESG variables themselves have a high correlation; for example, the correlation of $ESGC_{i,t-1}$ and $ESGC_{i,t}$ is 0.79; $ESGC_{i,t}$ and its elements $ENV_{i,t}$, $SOC_{i,t}$, $GOV_{i,t}$ also have a high correlation. Therefore, these variables should not be used together in the same equation.

Appendix F. Time-effect test

	All firms		Large firms		SMEs	
	F	P-value	F	P-value	F	P-value
Model 1	7.54	0.00 ***	7.58	0.00 ***	1.88	0.11
Model 2	7.57	0.00 ***	7.67	0.00 ***	1.73	0.14
Model 3	7.14	0.00 ***	7.39	0.00 ***	1.34	0.25
Model 4	7.17	0.00 ***	7.98	0.00 ***	1.53	0.19

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All models show a significant time-fixed effect, except for SMEs. Therefore, regression models for SMEs do not include time effects.

Appendix G. Modified Wald test for groupwise heteroskedasticity

	All firms		Large firms		SMEs	
	Chi2	P-value	Chi2	P-value	Chi2	P-value
Model 1	9.2e+34	0.00 ***	7.6e+30	0.00 ***	3.8e+34	0.00 ***
Model 2	6.3e+30	0.00 ***	1.9e+31	0.00 ***	9.6e+30	0.00 ***
Model 3	5.8e+31	0.00 ***	1.2e+31	0.00 ***	1.2e+34	0.00 ***
Model 4	3.8e+31	0.00 ***	1.9e+30	0.00 ***	5.1e+0.5	0.00 ***

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Heteroskedasticity is present in all models.

Appendix H. Wooldridge test for autocorrelation

	All firms		Large firms		SMEs	
	F	P-value	F	P-value	F	P-value
Model 1	7.99	0.01 **	4.50	0.04 *	4.20	0.04 *
Model 2	3.20	<i>0.07</i>	2.92	<i>0.09</i>	1.21	<i>0.27</i>
Model 3	7.29	0.01 **	3.88	0.05 *	4.11	0.04 *
Model 4	7.10	0.01 **	5.34	0.02 *	2.23	<i>0.13</i>

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The insignificant p-value is italicized. All models suffer from autocorrelation, except model 2 for all firms and model 4 for SMEs.

Appendix I. Pesaran test for cross-sectional dependence

	All firms		Large firms		SMEs	
	Pesaran	P-value	Pesaran	P-value	Pesaran	P-value
Model 1	2.18	0.03 *	1.98	0.05 *	-0.39	<i>0.70</i>
Model 2	2.55	0.01 *	2.38	0.02 *	-0.06	<i>0.96</i>
Model 3	2.42	0.02 *	2.05	0.04 *	-0.15	<i>0.88</i>
Model 4	2.33	0.02 *	1.94	0.05 *	-0.14	<i>0.89</i>

Note: Model 1: Regression with aggregated $ESGC_{i,t-1}$; Model 2: Regression with disaggregated $ENV_{i,t-1}$; Model 3: Regression with disaggregated $SOC_{i,t-1}$; Model 4: Regression with disaggregated $GOV_{i,t-1}$. Superscripts indicate significance level * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The insignificant p-value is italicized. Cross-sectional dependence is present in all models except for SMEs.



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